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How US Bank Regulation Failed SVB and Its Supervisors

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How US Bank Regulation Failed SVB and Its Supervisors¹

Greg Feldberg,² Carey K. Mott,³ and Jill Cetina⁴

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Abstract

It is well known that Silicon Valley Bank (SVB) failed in March 2023 because of a toxic combination of uninsured deposits and underwater securities. This article argues that the bank's failure could have been avoided if SVB had been subject to two global standards established by the Basel Committee on Banking Supervision. First, the interest-rate risk in the banking book (IRR-BB) standard, never fully implemented in the United States, would have identified the bank's extremely risky asset-liability management strategy and required remedial action 10 quarters before it failed. Second, the liquidity coverage ratio (LCR), from which US regulators had exempted banks of SVB's size, would have required the bank to hold tens of billions of dollars more liquidity at least four quarters before it failed. The article also argues that the Federal Deposit Insurance Corporation would have saved USD 13.6 billion if SVB and other banks that failed in 2023 had been subject to the Basel Committee's total loss-absorbing capacity (TLAC) standard, which requires banks to issue bonds that can be converted to equity. SVB's failure resulted in official sector protection of uninsured depositors through the US authorities' invoking of the systemic risk exception and the Federal Reserve's creation of the Bank Term Funding Program. Given public expectation of government and central bank policy intervention in a crisis and elevated levels of US public sector debt, bank supervision must emphasize ex ante crisis prevention. The conclusion is that US bank regulation needs to include quantitative limits on interest-rate risk, liquidity risk, and TLAC for more US banks—not just the systemic US banks. Such rules are particularly important given ongoing risks related to still above-target inflation, Treasury market fragility, growing risks to investor confidence in Treasuries, and the Fed's use of unconventional monetary policy, which increases the volatility of both US banking system liquidity and uninsured deposits.

Keywords: bank regulation, bank supervision, interest-rate risk, liquidity coverage ratio, Silicon Valley Bank, total loss-absorbing capacity

¹ This note builds on five blog posts that were originally published on the Yale School of Management's Systemic Risk blog between March and July 2023.

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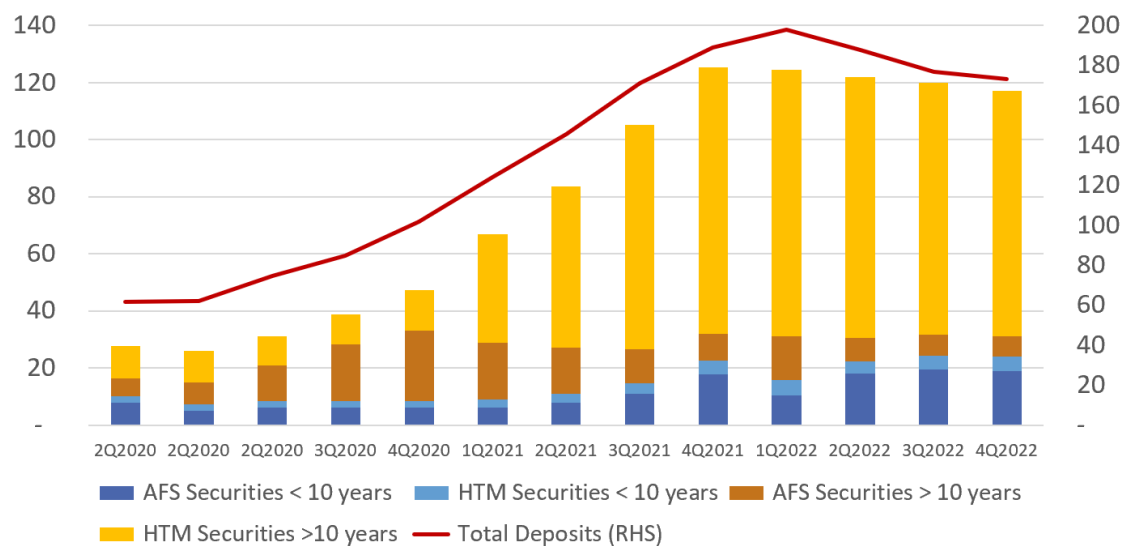
I. Introduction

Silicon Valley Bank (SVB) was the first big US bank to fail since the Global Financial Crisis of 2007–09. The two factors that coincided for SVB and caused the run on the bank were its extraordinary overreliance on deposits that were not insured by the Federal Deposit Insurance Corporation (FDIC), on the one hand, and the significant amount of unrealized losses the bank had on long-term securities, on the other. When the bank announced a stock sale to raise funds to cover realized losses on some securities, the market quickly turned its attention to the much larger unrealized losses on the bank's books and uninsured depositors began to worry that their money was at risk. Within two days, the bank experienced an extraordinarily swift deposit run and regulators shut it down.

SVB's Bad Gamble, in Short

SVB sowed the seeds of its failure during the COVID-19 crisis, when massive government fiscal stimulus and Federal Reserve quantitative easing created a surge in reserves at the Fed and uninsured deposits for the entire banking system. The venture capital industry, SVB's core clientele, boomed, and SVB's management saw a profitable trade—they took those low-interest deposits from clients and invested in residential mortgage-backed securities (MBS) with maturities of more than 10 years. Deposits rose 65% in 2020 and 85% in 2021; SVB's holdings of residential mortgage-backed securities rose 90% in 2020 and 250% in 2021. Per Figure 1, most of SVB's securities holdings were mortgage-backed securities, classified as held-to-maturity (HTM), with maturities of more than 10 years. It also had a smaller portfolio of available-for-sale (AFS) securities, mostly Treasuries with maturities of less than 10 years (Figure 1).

Figure 1: SVB's Gamble on Long-Term Securities (USD billions)



Sources: Company financial reports; YPFS analysis.

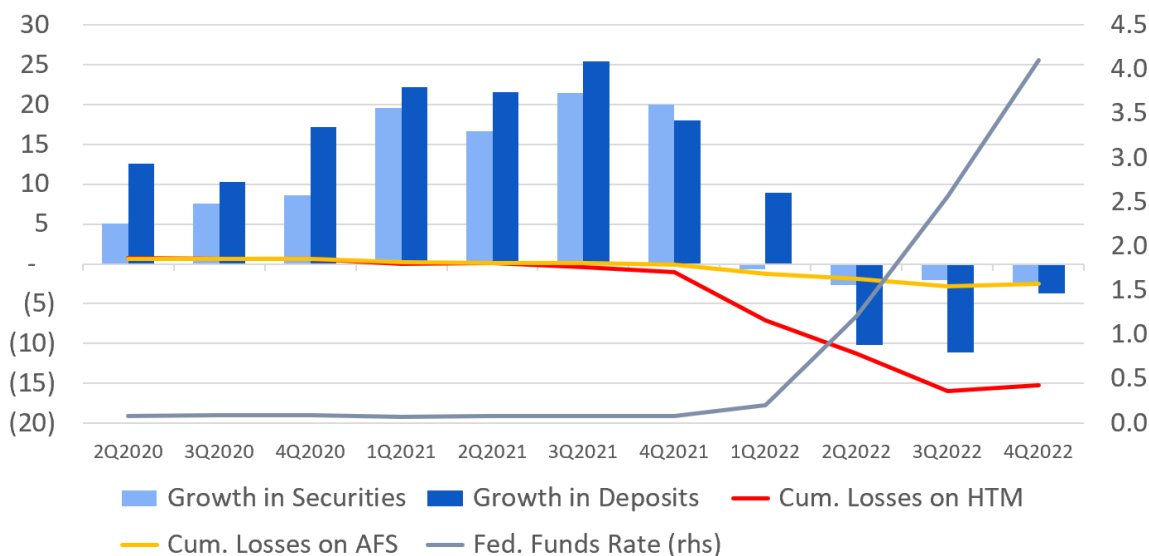
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For a while, the bank's investment strategy was profitable, and the equity market celebrated SVB. One widely read financial blog published an article in May 2021 titled "SVB Financial Group is the Best Bank Stock Nobody's Talking About." The author noted the bank's stock had more than doubled over the past year, and went on to say:

A best-ever quarter in the first period of 2021 underpins the stock's performance, but this story is far from over. The future remains bright for this banker of the 'innovation economy.' . . . Basically, even though average interest rates are far lower now than they were pre-pandemic, Silicon Valley Bank's assets and loans have grown so much they're still netting more in interest income despite the low rate environment. (Rossolillo 2021)

SVB's stock continued to rise in 2021, peaking at more than four times its 2020 low. But the party was soon over. In response to significantly above-target inflation, the Fed raised the benchmark federal funds rate rapidly in 2022, resulting in mark-to-market losses on SVB's huge portfolios of negatively convex long-term mortgage-backed securities and US Treasuries. By the end of the year, it had USD 15 billion in unrealized losses on those securities, almost equal to its entire equity capital (Figure 2). The venture capital industry also slowed down, leading SVB's customers to withdraw cash. SVB's deposits fell 9%, putting net interest income under pressure, as the bank replaced lost deposits with more expensive wholesale funding.

Figure 2: SVB's Accumulating Unrealized Losses (USD billions)



Sources: Company financial reports; YPFS analysis.

Of course, mark-to-market losses on held-to-maturity securities do not have to be debilitating if a bank has sufficient other liquidity and is able to hold those securities to maturity, implying that its assets and liabilities are well matched. But in SVB's case, they weren't. In a few days in March 2023, the bank fell apart rapidly. On March 8, SVB announced it had sold USD 21 billion of US Treasuries that had been booked as available-for-sale,

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virtually its entire AFS portfolio—presumably to meet pressing liquidity needs—and that it planned to raise USD 2.25 billion in new capital to cover the realized losses on those sales.

There would be no buyers for that equity raise. The stark reality that the bank now had no further options to raise liquidity than to sell held-to-maturity securities, which would result in the immediate realization of much larger losses, sent the bank in a tailspin. SVB had not pledged those securities to use as collateral at the Fed's discount window or with lenders in the repo, or repurchase agreement, market. Still, it is unlikely that this would have been sufficient to cover the bank's growing depositor demands. SVB depositors withdrew USD 42 billion on March 9 and sought another USD 100 billion the next morning, Friday, March 10. Regulators took over the bank during the day on Friday, an extraordinary move, but unavoidable given the circumstances.

With the full benefit of hindsight, the bank's failure appears inevitable. Most of the securities it owned paid fixed rates and had maturities of more than 10 years. It was simple financial math that they would lose value quickly if inflation got out of hand and the Fed raised interest rates. Yet, the bank's management didn't think that would happen. They expressed the view during 2021 and 2022 that they were prepared for further Fed rate cuts, not rate hikes. Their portfolio reflected a huge one-way bet on lower rates. SVB also believed their newfound deposits were stable.

The Thought Experiment: Prudential Rules in Hindsight

This paper asks whether certain rules, if they had applied to SVB, could have prevented SVB's management from taking that huge one-way bet on interest rates.

It is clear from Figures 1 and 2 that SVB's entire securities portfolio was locked in by the second half of 2021, a year and a half before its failure, as was its heavy reliance on uninsured deposits for funding. Sometime in 2021, the bank could have prevented its eventual demise by taking steps to avoid or mitigate those risk concentrations. For example, it could have declined deposits to avoid further rapid balance sheet growth. It already managed a substantial amount of its customers' cash off its own balance sheet, in money market funds. The managed cash was roughly equal to the bank's on-balance-sheet, uninsured deposits. To avoid taking excessive risks with uninsured deposits, management could have simply decided to place much more of its customers' cash off its balance sheet, in money market funds, rather than accepting billions of dollars of deposits on its balance sheet. Alternatively, it could have held these balances at the Fed or in much shorter-term securities to avoid such a large exposure to rising interest rates.

Either strategy would have resulted in lower profits in the short term. But it is the role of bank management and the board to prudently manage the trade-off between short-term profits and long-term viability. It is also the role of bank supervisors to identify excessive risk-taking that could compromise a bank's safety and soundness. Thus, a key focus of the public discussion after SVB and two other large US banks failed in 2023 was whether weaknesses in supervision or regulation had been part of the problem.

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We focus our analysis on a discrete and arguably answerable question: If certain regulations had been in place and enforced, would they have made a difference? Could these regulations have prevented the dramatic risk management failures at SVB and related failures of supervisory oversight if regulators had applied them to the bank two or three years earlier? More specifically, could they have prevented the massive accumulation of uninsured deposits and held-to-maturity securities on SVB's balance sheet?

Our conclusion, with the obvious benefit of hindsight, is that better prudential regulation could have made a difference. We focus on two global standards established by the Basel Committee on Banking Supervision (BCBS) that were not applied to SVB or the other US banks that failed in 2023, Signature and First Republic:

- In Section II, we show that the interest-rate risk in the banking book (IRR-BB) standard, never fully implemented in the United States, would have identified SVB's extremely risky asset-liability management (ALM) strategy and required remedial action 10 quarters before it failed.
- In Section III, we show that the liquidity coverage ratio (LCR), from which US regulators had exempted banks of SVB's size, would have required the bank to hold tens of billions of dollars more cash at least four quarters before it failed.

This article also shows, in Section IV, that SVB would have appeared sound under the other Basel liquidity standard, the net stable funding ratio (NSFR). We argue that this result reflects potential weaknesses in the NSFR standard.

Section V argues that the FDIC would have saved USD 13.6 billion if SVB and other banks that failed in 2023 had been subject to the Basel Committee's total loss-absorbing capacity (TLAC) standard, which requires banks to issue debt that can be bailed in to support the bank's resolution.

II. Interest-Rate Risk in the Banking Book

The transformation of depositors' savings into investments is foundational to banking and permits higher levels of economic growth. But that transformation also creates challenges for asset-liability management owing to the typical maturity mismatch between short-term deposits and longer-term loans and securities. Changes in the level of interest rates can affect a bank's profitability and economic value in complex ways. Poor management of that interest-rate risk has been at the root of past financial crises, most notably the US savings and loan (S&L) crisis of the early 1980s.

US Reluctance to Implement the Basel Standard

In 2004, the Basel Committee on Banking Supervision developed a global standard for the management of interest-rate risk in the banking book (BCBS 2004). It strengthened that

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standard in 2016 on the expectation that interest rates, then near all-time lows, would eventually rise (BCBS 2016a).

The BCBS IRR-BB standard uses measures of the impact of potential changes in interest rates on a bank's net interest income (NII) and economic value of equity (EVE). EVE is a long-term metric that computes the net present value of a bank's total assets minus the net present value of total liabilities, including deposits.

The current Basel standard includes five key features:

- A framework for banks' EVE calculation, including caps on the quantity of deposits they can assume to be core and caps on the average assumed maturity by deposit category.
- A predefined set of six interest-rate-shock scenarios.
- A test that establishes that a bank is an "outlier" if the worst impact of those shocks on its EVE would exceed 15% of Tier 1 capital.
- An expectation that supervisors will require outlier banks to hold more capital or take other mitigating actions.
- Public disclosure of banks' interest-rate sensitivity measures.

Other jurisdictions, such as Canada, the UK, and the euro area, published rules to implement the Basel IRR-BB standard, but US regulators never did. Way back in 1991, Congress specifically directed US regulators to impose a mandatory capital charge to address interest-rate risks in US banks, in response to lessons learned from the S&L crisis. In 1992, regulators did propose a capital charge for outliers based on a simpler test of economic value. But they retreated in 1996 in response to industry pressure and never revisited the proposal.

Why the reluctance? US regulators said in 1996 that their decision "not to pursue a standardized measure and explicit capital charge for interest-rate risk at this time . . . reflects concerns about the burden, accuracy, and complexity of a standardized measure and recognition that industry techniques for measuring interest-rate risk are continuing to evolve" (Fed 1996).

The quarterly filings that bank regulators require US banks to publish do not include NII or a standard EVE calculation for the six BCBS interest-rate risk scenarios. However, under a 1997 rule, the Securities and Exchange Commission (SEC) does require all publicly listed US banks to disclose their sensitivity to interest-rate shocks in their annual Form 10-K filings (US SEC 1997). However, there is no uniformity in how US banks compute these disclosed interest-rate sensitivities.

Given the US experience with the S&L crisis, which was set off by the Fed's sharp tightening cycle under Chair Paul Volcker, US regulators should have taken interest-rate risk more

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seriously. The Office of Financial Research, an independent data and research arm of the US Treasury, flagged potential ALM risks facing US banks in its 2014 annual report (OFR 2015). This report highlighted:

[A] reversal of banking system deposits is a potential risk once interest rates rise. U.S. banks have seen dramatic growth in their non-interest-bearing deposits relative to total banking system liabilities. There is a non-negligible risk that these deposits could shift to alternative, higher-yielding investments as rates rise.

The report also emphasized the potential role of supervisory stress tests in addressing such risks:

Supervisory stress tests currently assume credit losses are the driver that will pressure capital. It is also possible that the driver of a bank's stress is not a macroeconomic shock, but a funding stress that arises inside the financial sector or a credit shock that could be magnified by funding stress. Potential liquidity and solvency interactions receive little consideration. For example, banks have increased securities holdings in held-to-maturity portfolios by nearly half a trillion dollars since the crisis. Under stress, banks may need to use these securities to generate liquidity, either by selling them (which could result in a capital loss) or by financing them through repos (which could reduce a bank's leverage ratio and LCR).

Subsequently, in 2015, one of the authors of this article expanded on these ideas by examining how four distinct types of shocks—funding, credit, liquidity, and interest rate risk/collateral—could interact to impact banks' capital and liquidity ratios. This analysis further noted that US supervisory stress tests at the time did not account for mark-to-market shocks on banks' growing securities portfolios, potentially leaving vulnerabilities unaddressed (Cetina 2015).

Yet, for over a decade, as highlighted in a Peterson Institute blog post-SVB, no US supervisory stress test has considered rising interest rates (Honohan 2023). Instead, US supervisory stress tests have focused on banks' potential trading book and loan losses. Thus, in the US, there are no incremental capital requirements for large US banks with high levels of interest-rate risk.

Limited Guidance on IRR-BB in Examination Manuals

The US has implemented some elements of the Basel IRR-BB standard for the largest banks through its supervisory processes, rather than through regulation. For example, the examination Handbook of the Office of the Comptroller of the Currency (OCC), the supervisor of banks with national charters, advises examiners to make sure the largest banks test their NII and EVE against the Basel Committee's six prescribed interest-rate scenarios. However, neither the OCC nor any other US regulator requires banks to make these calculations in a standardized manner (OCC 2020).

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Interestingly, the previous edition of the Handbook stated that for banks with more than USD 250 billion in assets, an EVE reduction in any scenario that exceeded 15% of the bank's Tier 1 capital would be considered an outlier, and an examiner should take that into account in forming conclusions about the bank's risk management. However, in 2019, the OCC revised the scope of its guidance on interest-rate risk as part of the regulatory tailoring initiative mandated by Congress so that it applies only to banks with more than USD 700 billion in assets (Fed, FDIC, and OCC 2019). Banks with between USD 250 billion and USD 700 billion in assets are still required to run the six Basel scenarios, but the OCC Handbook no longer applies the 15% outlier test or describes consequences for an examiner's evaluation of risk management for banks of that size. SVB never had more than USD 250 billion in assets.

The Federal Reserve's Commercial Bank Examination Manual is even more limited in its discussion of interest rate risk and states that Fed examiners may focus only on a bank's compliance with "self-imposed parameters" (Fed 2023b). Based on the Federal Reserve's manual, it can be difficult for a bank supervisor to challenge a bank's level of interest-rate risk as excessive if the bank is operating within the NII and EVE risk limits that the bank's board approved.

Limited Focus on IRR-BB in Supervisory Ratings

The supervisory ratings framework for banks in the US includes a rating for the sensitivity of a bank's balance sheet to market risk—including interest-rate risk. It is the "S" in the "CAMELS" rating.⁵ However, the composite US bank supervisory CAMELS ratings are not simply an average of the individual component ratings but instead reflect examiners' judgment as to how the individual components ratings are combined to provide a summary measure of a bank's overall financial condition. Thus, in theory, a poor S rating—which includes a bank's interest-rate risk—could drive an institution's composite CAMELS rating down and, in turn, have meaningful real-world ramifications for a supervised entity. For example, an institution with a "4" CAMELS composite rating means that close supervisory attention is required and, in most cases, a formal enforcement action may be necessary.

However, other research suggests that S ratings have had little impact on banks' CAMELS composite ratings. Gopalan and Granja (2023) find that composite CAMELS ratings were not significantly affected by exposures to interest-rate risks even after the Federal Reserve began raising interest rates in 2022. Similarly, Gaul and Jones (2021) find in their analysis of US bank supervisory data from 1984 to 2020 that only about 1% of the variation in a bank's CAMELS composite rating is attributable to the variation in the bank's S rating. This result seems quite low for interest-rate risk, as maturity transformation is a foundational banking risk. It also suggests that the absence of the US implementation of thresholds for excessive interest rate per the Basel IRR-BB standard risk may limit effective US bank supervision of this risk.

⁵ US bank examiners use the CAMELS ratings framework to evaluate bank safety and soundness. It includes a composite measure, on a five-point scale, and six subcomponents, also on a five-point scale: capital adequacy, asset quality, management, earnings, liquidity, and sensitivity to market risk.

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In sum, publicly listed US banks do face SEC requirements to annually disclose their interest-rate sensitivity but not to publish these metrics as part of their quarterly regulatory filings for bank regulators. Moreover, these metrics and their calculation are not standardized per the BCBS methodology, there are no supervisory thresholds with consequences, and US banks with high levels of interest-rate risk are rarely subject to having their composite supervisory rating downgraded.

Thought Experiment: IRR-BB as a Warning Signal for SVB

Data from SVB's 10-K filings suggest that the Basel Committee's outlier test for interest-rate risk would have flagged Silicon Valley Bank in four out of the five years between 2017 and 2021 if US regulators had held the bank to that standard (Figure 3). The bank's EVE losses would have been unacceptably high under a falling-rate scenario in 2017 and 2018 and unacceptably high under a rising-rate scenario in 2020, 2021, and the first three quarters of 2022. SVB stopped publicly reporting this information in the fourth quarter of 2022.

Figure 3: SVB's Economic Value of Equity (EVE) Sensitivity, 2017–21

SVB Financial Model Results					
\$ billions					
	2017	2018	2019	2020	2021
Base Economic Value of Equity	7.3	8.8	8.1	11.2	20.7
EVE at stressed +200 bps	8.1	9.3	7.5	9.5	15.0
EVE decline at stressed +200 bps	N/A	N/A	(0.6)	(1.7)	(5.7)
EVE decline in percent	11%	6%	-8%	-15%	-28%
EVE at stressed -200 bps	5.7	7.6	8.6	11.5	21.3
EVE decline at stressed -200 bps	(1.6)	(1.3)	N/A	N/A	N/A
EVE decline in percent	-22%	-14%	5%	3%	3%

SVB Balance Sheet					
Total Capital	4.6	5.6	6.6	8.2	16.7
Tier 1 Capital	4.2	5.2	6.3	7.7	16.2
Basel 2016 threshold: 15% of Tier 1 Capital	0.6	0.8	0.9	1.2	2.4
Basel pre-2016 threshold: 20% of Total Capital	0.9	1.1	1.3	1.6	3.3
Max EVE decline under stress	(1.6)	(1.3)	(0.6)	(1.7)	(5.7)
Max EVE decline: Percent of Tier 1 Capital	-37%	-24%	-10%	-22%	-35%
Max EVE decline: Percent of Total Capital	-34%	-23%	-9%	-21%	-34%
EVE decline exceeds Basel 2016 threshold?	YES	YES	NO	YES	YES
EVE decline exceeds pre-Basel 2016 threshold?	YES	YES	NO	YES	YES
Capital shortfall: Basel 2016 standard	(0.9)	(0.5)	0.3	(0.6)	(3.3)

Sources: Company financial reports; YPFS analysis.

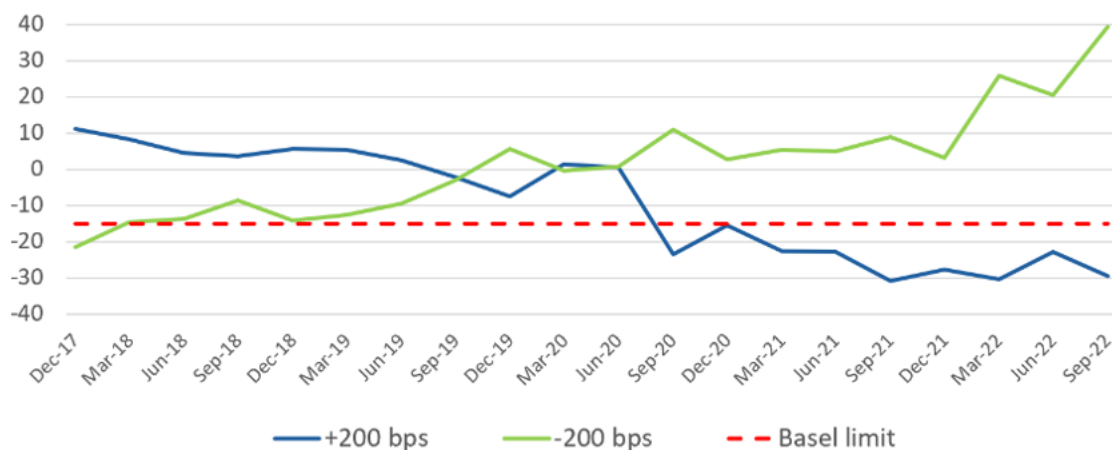
In September 2020, the IRR-BB EVE metric immediately pointed to trouble under rising-rate scenarios when SVB started piling new deposits into long-term securities. Figures 4 and 5 show the impact of 200-basis point (bp) and 100-bp parallel shocks on SVB's EVE over time, including a falling-rate scenario (green lines) and a rising-rate scenario (blue lines), based on SVB's 10-K and 10-Q filings. Figure 4 shows that the modeled declines in EVE would have exceeded the Basel threshold, represented by the dotted red line, starting in the third quarter of 2020. Figure 5 shows that the EVE declines also would have exceeded the bank's internal

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“outer” limit, which its board set at 10% of EVE in a 100-bp parallel shock, in the third quarter of 2020. SVB would have come back into compliance with the internal limit in the fourth quarter but gone out of compliance again throughout 2021 and 2022. Of course, in 2022, the Fed raised interest rates not 100 bps or 200 bps, but 400 bps.

Meanwhile, there is no doubt SVB was an outlier when compared to other US banks. The median OCC-supervised bank with more than USD 10 billion in assets would have seen a 5% *increase* in its EVE in the face of a 200-bp shock in the 21-month period through June 2021, according to the Office of the Comptroller of the Currency’s semiannual Interest Rate Risk Statistics Reports, which are based on statistics gathered from hundreds of banks recently examined by the OCC (OCC 2024). The 25th-percentile bank loss was just 1%, and the highest estimated loss for any bank in the survey was 17%. SVB’s estimated EVE losses in both the 100-bp and 200-bp rising-rate scenarios were consistently worse than the worst large bank in the OCC’s survey from September 2020 until the bank stopped publicly reporting this information, with the exception of the blip in December 2020 (Figure 6).

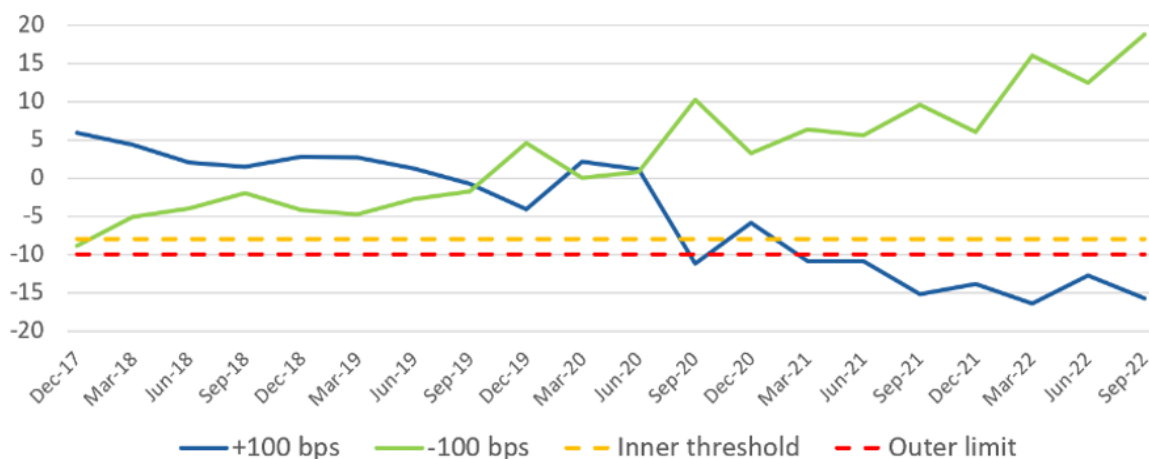
Figure 4: Impact of 200-bp Parallel Shocks on SVB’s EVE vs. Basel Threshold (% EVE)



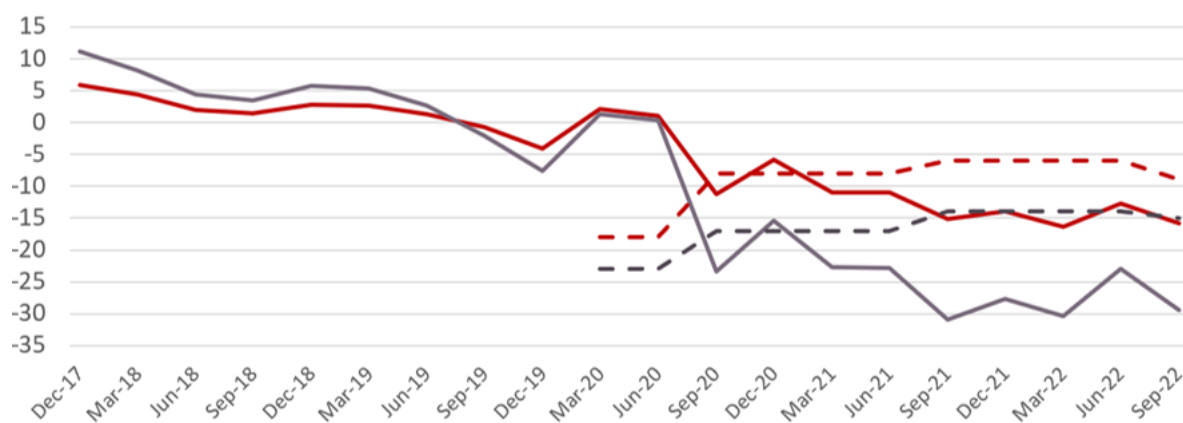
Note: SEC reporting shows changes in estimated EVE; Basel shows Tier 1 equity.

Sources: Company Forms 10-K and 10-Q.

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Figure 5: Impact of 100-bp Parallel Shocks on SVB's EVE vs. Internal Limits (% EVE)

Sources: Company Forms 10-K and 10-Q.

Figure 6: SVB's EVE Sensitivity vs. the Worst Bank with More than USD 10 billion in Assets in the OCC's Biannual Survey (% EVE)

Sources: Company Forms 10-K and 10-Q; OCC statistical reports; authors' analysis.

IRR-BB Conclusions

The Federal Reserve's May 2023 report on SVB's failure targeted its supervisors for not demanding that the bank address its interest-rate risk deficiencies:

SVBFG management was focused on the short-run impact on profits . . . With regard to interest-rate risk-management, supervisors identified interest-rate risk deficiencies in the 2020, 2021, and 2022 CAMELS exams but did not issue supervisory findings (MRA/MRIA). The deficiencies were only communicated as written advisories or verbal observations. (Fed 2023a)

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On the other hand, examiners are typically at a disadvantage when challenging management of a profitable bank about risks that may or may not manifest. A specific, quantitative regulatory standard with an outlier test, like the Basel IRR-BB standard, would have been a useful tool for SVB's supervisors in 2017–18 and again in 2020–22. The Basel outlier standard, backed up by comprehensive regulatory data on interest rate risk, would have provided solid evidence that SVB was not operating in a safe and sound manner. Strong rules can both empower competent supervisors and act as a floor on bank management and supervisory incompetence.

III. Liquidity Coverage Ratio

The liquidity coverage ratio is a Basel standard introduced after the Global Financial Crisis. It requires banks to hold sufficient high-quality liquid assets (HQLAs) to manage expected net cash outflows in a 30-day stress scenario.

In the original version of the rule implementing the Basel LCR standard that US regulators finalized in 2014, banks with USD 250 billion in assets or USD 10 billion in foreign exposures had to maintain their LCR ratios above 100%. SVB would have been subject to that ratio because its foreign exposures met the threshold.

However, in 2019, the US bank regulators, following an act of Congress, determined that most banks with between USD 50 billion and USD 250 billion in assets would no longer be subject to the LCR or other enhanced prudential standards applied to the most systemically important banks (Fed, FDIC, and OCC 2019; Fed 2019). As a result of this regulatory tailoring in the US, banks of SVB's size and smaller and banks without a bank holding company (BHC) were no longer subject to the LCR.

Thought Experiment: LCR as a Warning Signal for SVB

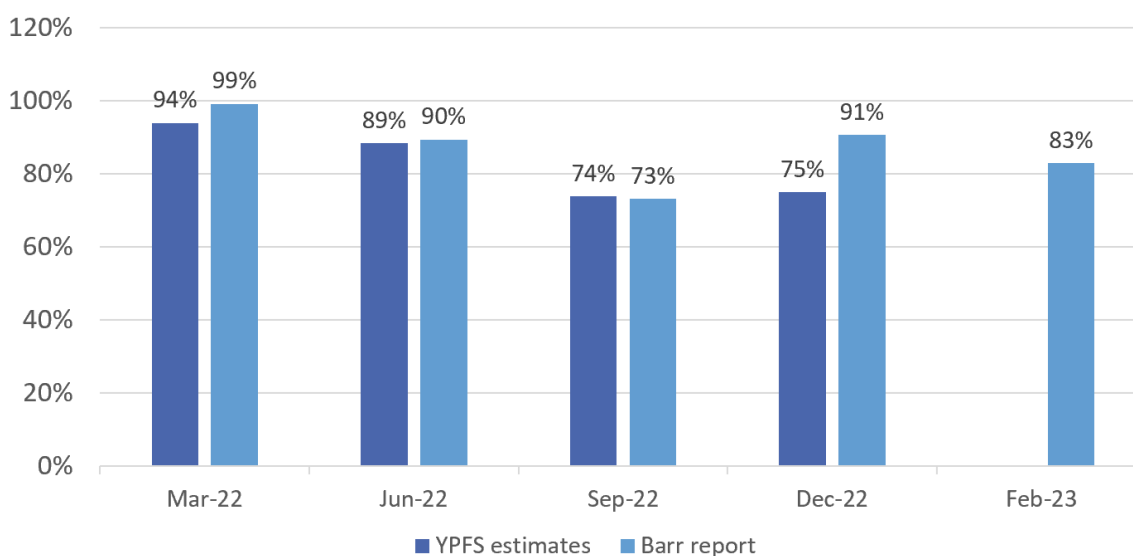
SVB's first quarter 2023 public financials suggest that its LCR would have been 75% at the end of 2022, substantially below the 100% threshold, if regulators had required the bank to report it. See our calculations in Appendix A. The Federal Reserve's material loss review of the SVB failure also noted that SVB's supervisors did not believe that SVB's liquidity buffer was adequate. One clear conclusion from our analysis is that it is a challenge to calculate the LCR working with the available public data for a company that is not required to disclose LCR data. Given SVB's undercapitalization, a more solid liquidity buffer could have reassured uninsured depositors.

But the bank would have long been out of compliance with the LCR standard. On April 28, 2023, the Federal Reserve issued Vice Chair Michael Barr's report on the failure of SVB (Fed 2023a). The report, on page 88, included the Fed's calculation of SVB's LCR on a monthly basis from March 2022 to February 2023, based on confidential data available only to the Fed. Over those 12 months, the report said, the bank's LCR never exceeded 100%, ranging from a high of 99% in March 2022 to a low of 73% in September 2022.

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Figure 7 represents an admittedly rough effort to fit our methodology to the quarter-end data in the Barr report. Figure 8 takes those figures and estimates how much additional HQLA the company would have needed to meet a 100% LCR requirement. Figure 9 shows the HQLA gaps assuming a 125% LCR, including the typical management buffer that US global systemically important banks (G-SIBs) that are subject to the rule tend to maintain. (See BCBS [2025] for a dashboard of updated LCR averages on a regional basis.) The conclusion: If subject to the LCR, SVB would have needed to take actions to increase its HQLA by USD 20 billion more than a year before it failed and would have been flagged as having a USD 40 billion HQLA deficiency six months before it failed.

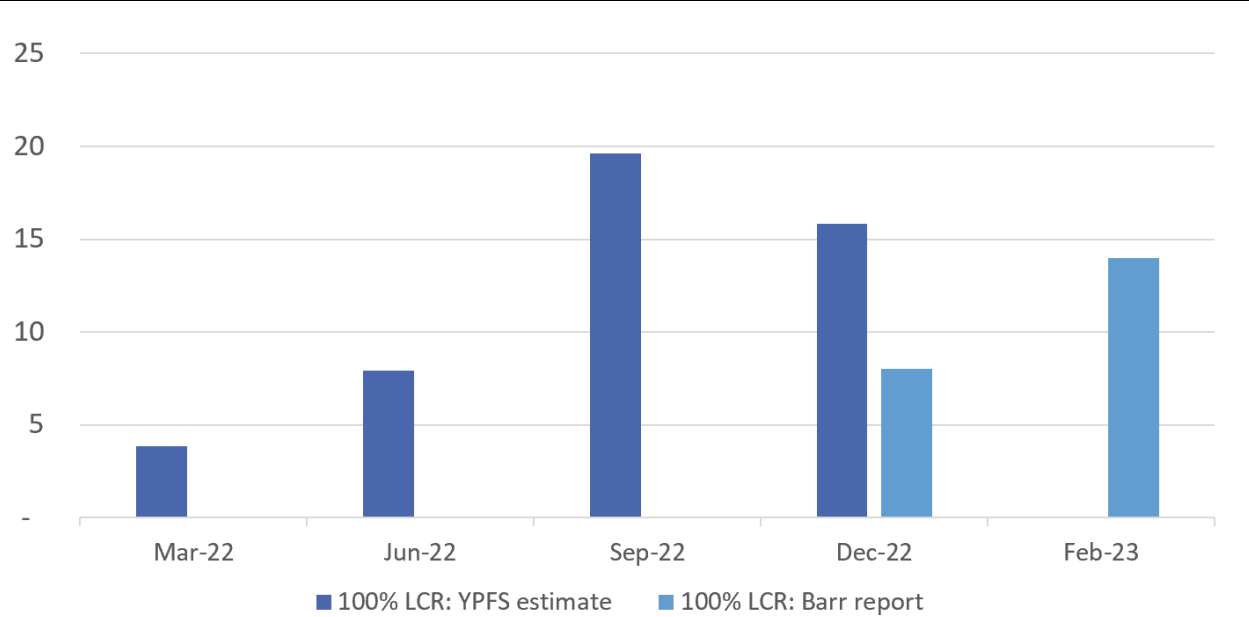
Figure 7: SVB's LCR over Time



Sources: Company financial reports; Fed 2023a, 88; YPFS analysis.

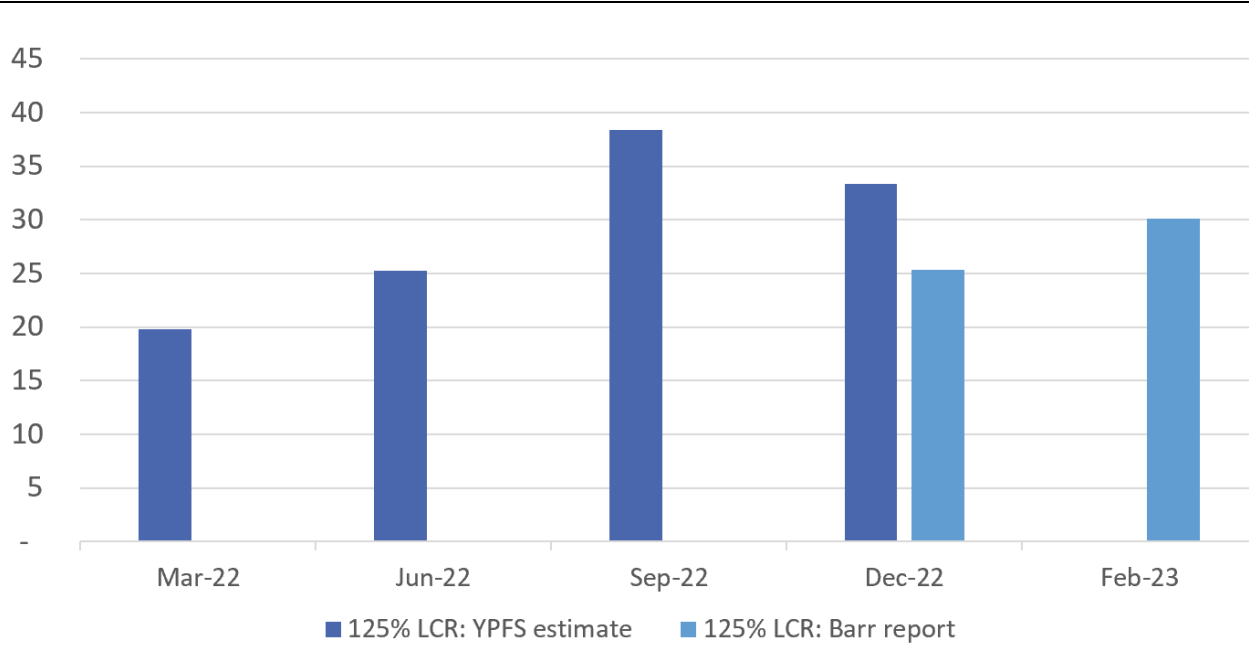
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Figure 8: SVB’s HQLA Shortfalls vs. 100% LCR



Sources: Company financial reports; Fed 2023a, 88; YPFS analysis.

Figure 9: SVB’s HQLA Shortfalls vs. 125% LCR



Sources: Company financial reports; Fed 2023a, 88; YPFS analysis.

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Of course, if the bank were subject to the rule, its supervisors would not have allowed its LCR to fall so far or allow its HQLA buffer to become so concentrated in negatively convex agency MBS due to the LCR's limits on these securities. Bank management and supervisors shouldn't just react when a bank breaches a limit—they should act when limit breaches become foreseeable. The LCR rule also brings additional supervisory oversight. As the Barr report noted:

The LCR rule also requires a firm to have the operational capability to monetize its liquid assets and to test this capability periodically. In addition, the LCR rule places limits on the composition of assets that qualify as high-quality liquid assets. If SVBFG had been subject to the LCR, it may have adopted more proactive monitoring or managing of its liquidity position and mix of liquid assets. (Fed 2023a, 88)

On the other hand, if subject to the rule, SVB may have chosen to improve its LCR through other means, in addition to holding more HQLAs. It may have followed a more responsible funding strategy, with more long-term debt (LTD) and less uninsured deposits, as the bank itself said in its 2022 10-K, published less than a month before it failed:

It is possible that, as a result of further growth, we may . . . become subject to LCR and NSFR requirements or other heightened liquidity requirements in the future, which would require us to maintain high-quality liquid assets in accordance with specific quantitative requirements and increase the use of long-term debt as a funding source. (SVB 2023a, 87)

The bulk of SVB's unrealized securities losses were in its HTM portfolio. US GAAP does not permit any interest-rate derivatives to qualify for hedge accounting for HTM securities. Thus, because the mark-to-market gains and losses for any interest-rate derivatives would pass directly through a bank's income statement, US banks do not hedge HTM securities. This can result in bank management poorly managing the interest-rate risk associated with HTM securities. However, the US LCR rule requires using market valuations for HTM securities for purposes of measuring a bank's HQLA. Thus, application of the LCR to SVB could have helped reduce the outsized interest-rate risk in its HTM portfolio. Specifically, SVB's unrealized HTM losses would have required it to take remedial action to strengthen its liquidity buffer or disincentivized such behavior.

Moreover, if the bank had been subject to the LCR, it would have been required to publicize more data about its liquidity risks. The market, not just bank supervisors, may have been more focused on the risks it faced. Uncertainty about both solvency and liquidity and the ability to recover one's deposit in full is an essential element of a bank run.

To be clear, no LCR-covered bank could have survived the run SVB faced in March 2023, but SVB was not covered by the LCR. The role of the LCR is not to allow a bank to manage any potential crisis but to promote sound liquidity risk management to help prevent it from facing a crisis.

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LCR Conclusions

SVB's failure carries an important lesson about the liquidity of underwater long-term assets and discount window readiness. In contrast to SVB, large banks in the US that were subject to the LCR managed the spring 2023 crisis in part because they were subject to heightened oversight and had built substantial liquidity buffers. While some LCR-covered US banks had substantial unrealized securities losses due to rising interest rates, the LCR standard did force these banks to increase their HQLAs before 2023. Cipriani, Eisenbach, and Kovner (2024) analyzes bank runs using Fedwire data and finds that no US banks with assets more than USD 250 billion experienced runs. The authors conclude that the absence of runs for these banks in the spring of 2023 is evidence of a too-big-to-fail effect. However, their analysis focuses solely on bank balance sheet characteristics and does not test for regulatory differences as a possible explanatory variable. Given SVB's and First Republic's assets—more than USD 200 billion but less than USD 250 billion—the paper's findings leave open the possibility that the 2019 narrowed scope of application of the US LCR rule due to regulatory tailoring may have also played a role.

There is broad agreement in the US that most banks are small and should not need to follow enhanced prudential standards designed for large, complex, and internationally active banks. But, with more than USD 200 billion in assets, SVB was not small. Meanwhile, the US remains unusual among developed nations in applying the LCR only to the very largest banks. Most jurisdictions—the UK, Europe, most of Asia—have maintained full LCR disclosure and threshold requirements for most or all banks. Yet, there has been little public discussion in the US since the 2023 banking failures about the possibility of extending LCR regulation to midsized US banks and the role of the 2018 regulatory tailoring in the 2023 US bank failures.

So, the key takeaway from the SVB failure is that banks like SVB should be subject to the LCR, not that the LCR itself failed or needs to be fundamentally reconsidered.

Still, some tweaks to the LCR's design may deserve consideration. Concentrated uninsured deposits could merit a higher runoff rate. It may also be appropriate to limit the duration of securities held in banks' HQLAs. The lesson specific to SVB is that HTM securities are not perfectly liquid and agency MBS are negatively convex and subject to significant extension risk. As deposits also contain an embedded option that allows the depositor to withdraw funds at any time, agency MBS funded by uninsured deposits is akin to a bank being doubly short interest-rate volatility. Liquidity is the ability to ensure the availability of funds to meet commitments at a reasonable price at all times without damaging capital. Finally, unrealized gains and losses on available-for-sale securities are incorporated in capital only for banks with more than USD 700 billion in assets, not for banks of SVB's size.

IV. Net Stable Funding Ratio

The net stable funding ratio is the other international standard introduced in the wake of the Global Financial Crisis to address banks' funding and liquidity risks. It measures banks'

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maturity transformation risk over one year—that is, the extent to which banks fund long-term assets with short-term liabilities. Maturity transformation is the core business of banking, of course, and it creates the most basic risks that banks must manage.

The NSFR requires banks' available stable funding (ASF) to exceed their required stable funding (RSF) over a 12-month time frame. The NSFR complements the LCR. While the LCR focuses on short-term liquidity risk, the NSFR focuses on a bank's structural asset-liability management.

In contrast to the LCR, SVB would have complied with the NSFR. We estimate the bank's NSFR would have been 132%, as explained in Appendix B (Figure 12). That would have put it comfortably above the regulatory 100% target and above the global average of about 120% (which excluded US banks until 2023; see BCBS [2025]). Based on that estimate, SVB's NSFR would not have raised any alarms about the bank's extraordinary exposures to uninsured deposits on the liability side of its balance sheet and long-dated agency mortgage-backed securities on the asset side.

However, that result does not mean that SVB managed its funding risks appropriately. Rather, the result reflects potential weaknesses in the 2020 NSFR final rule itself and in the 2014 Basel standard on which US regulators based the rule (BCBS 2014; Fed, FDIC, and OCC 2020). First, how stable should the rules consider uninsured wholesale deposits to be, given the runs in this episode and earlier episodes? Second, are the haircuts on HTM securities sufficient? Specifically, under the NSFR, if an HTM security qualifies as a HQLA, it will generally receive a lower required stable funding factor, meaning less stable funding is required to support its assumed high liquidity. Third, does it make sense to continue to exclude unrealized gains and losses on AFS securities from regional banks' capital measure, even as reported capital is assigned a 100% available-stable-funding factor under the NSFR? Note that no US banks were subject to the NSFR at the time of SVB's failure. The US NSFR rule required only the very largest US banks to begin publicly reporting their NSFRs in the second quarter of 2023. Even if the NSFR had been in place in 2023, SVB would have been exempted under the same tailoring rule that exempted banks of SVB's size and those without bank holding companies from the LCR (Fed, FDIC, and OCC 2019).

NSFR Conclusions

It is surprising that a bank like SVB taking extraordinary maturity transformation risks could have passed a regulatory metric designed specifically to identify such outliers. A few reasonable tweaks to the NSFR could have created more cautionary outcomes.

Most important is the treatment of uninsured wholesale deposits. The rapid growth of SVB's uninsured deposits was one of the biggest red flags the company's finances should have raised. Perhaps the rule could be revised to penalize newer uninsured wholesale deposits—give them a lower or even 0% ASF factor for the first year, for example. That would have helped flag rising risks in all fast-growing banks, not just in SVB.

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On the other hand, there were clearly idiosyncratic aspects of SVB's deposit base. The rule allows firms and their supervisors to adjust assumptions about funding factors in such cases, although it isn't clear how often supervisors do this. SVB's uninsured deposits were almost entirely from VC-funded companies. Many of these companies run negative cash flows every month and sustain their businesses through periodic injections of new capital. For that reason, considering those deposits to be 50% "available" over a 12-month horizon may seem extremely optimistic—and note that the rule considers such nonfinancial companies' deposits to be 50% available whether or not they are operational. Assuming 25% instead would have lowered SVB's NSFR in our model below 100%, all other assumptions remaining the same; assuming 0% would have lowered its NSFR to 66%.

The other issue that SVB's failure raises is the treatment of underwater HTM securities that qualify as HQLAs. The rule requires minimal available stable funding for such securities because it assumes an active repurchase agreement funding market. In other words, it assumes the repo market will fund 100% of Level 1 HQLAs and 85% of Level 2A HQLAs; it requires stable funding for the remainder—0% for Level 1 and 15% for Level 2A.

But if these securities have unrealized losses, the repo market (or other funding sources) won't fund their full carrying value. One tweak to the rule would require 100% stable funding for the difference between fair market value and carrying value for these securities. So, even Treasuries and high-quality agency mortgage-backed securities would require some stable funding in current market conditions, which would lead to unrealized losses. The rule could continue to require some stable funding based on collateral quality for the remaining fair value, to reflect the risk that funding may become expensive for such securities through higher haircuts.

Boosting the required stable funding in our model for Treasuries (Level 1) from 0% to, say, 15%—including both a 100% charge on a 10% loss and a small new 5% charge for the remaining fair value—and boosting the required stable funding for agency mortgage-backed securities (Level 2A) from 15% to 50%—including a 100% charge on a 20% loss and a higher 30% charge for the remaining fair value—would have lowered SVB's NSFR to 100%.

V. Total Loss-Absorbing Capacity⁶

The FDIC could have saved USD 13.6 billion in the resolution of SVB and two other large banks that failed in 2023 if regulators had earlier held the banks to the total loss-absorbing capacity standard, based on reasonable assumptions (see Appendix C).

In 2023, US regulators resolved three large, failed banks: SVB, Signature Bank, and First Republic Bank. The FDIC estimated then that coverage of uninsured depositors, discounts on

⁶ A version of this note was originally published on the Yale School of Management's *Systemic Risk* blog on July 6, 2023. The authors wish to thank Jordan Thursby, Paolo Colonna, and Stephanie Quiet for their helpful input and comments on the original version of this note. The views and analysis are those of the authors and remain personal to them.

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the purchase price of the banks, and potential losses on the assets it assumed would cost its Deposit Insurance Fund (DIF) a total of USD 31.5 billion, roughly half of which it recouped later from other large banks (Mitchell 2023).

Under the 2016 TLAC standard agreed by the Basel Committee on Banking Supervision, regulators began requiring the world's largest banks to maintain a minimum level of long-term, unsecured debt to complement their regulatory capital. Together, this debt and regulatory capital (equity) represent a bank's TLAC. If a bank's losses exceed its capital, the additional layer of debt can be "bailed in," that is, converted to equity, providing an additional cushion against the bank's insolvency.

The conversion of TLAC facilitates the open-bank resolution of bank holding companies. Regulators now expect the largest bank holding companies to have resolution plans under which they can enter bankruptcy without shutting their banking and investment banking subsidiaries. TLAC complements these plans because it recapitalizes these subsidiaries and keeps them open.

Currently, US regulators require all banks to meet minimum capital requirements. But they require only eight US global systemically important banks to meet the TLAC requirement, which includes the layer of long-term debt on top of capital. In late 2022, US regulators proposed extending the rule to the five or so additional domestic banks with greater than USD 250 billion in assets (Fed and FDIC 2022).

Following the 2023 bank failures—all three of which were below this USD 250 billion threshold—the banking regulators, in August 2023, issued a rule to subject banks with USD 100 billion or more in assets to long-term debt requirements comparable to those that G-SIBs face under TLAC; the proposed TLAC rule has yet to be finalized (Fed, FDIC, and OCC 2023).

The bank resolutions in 2023 offer an opportunity to study how a rule with a broader scope of application might have worked if it already had been in place.

Background on TLAC

Under the Basel TLAC standard, G-SIBs are required to hold an additional cushion on top of their minimum regulatory capital requirements (BCBS 2016b). In addition to common equity, TLAC includes unsecured debt with a remaining maturity of more than one year, so long as that debt is subordinated to deposits and other excluded liabilities. As of 2022, a G-SIB's total TLAC under the Basel standard should be the greater of (1) 18% of its risk-weighted assets (RWAs)—that is, the bank's loans, securities, and other assets, weighted to reflect the risk of loss to the bank—or (2) 6.75% of its leverage exposures—that is, the bank's total assets and off-balance-sheet exposures.

Beyond these minimum standards, implementation varies by country. The US applies a minimum 18% of RWA and 7.5% of leverage exposures—again, though, only to G-SIBs. The Basel standard also sets an expectation that LTD will form at least one-third of the TLAC

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requirement (see the standard, BCBS [2015, 3]). Following Basel, the US requires its G-SIBs to have 6% of RWA in LTD, or one-third of the 18% requirement.

Unlike the US, the EU requires banks of all asset sizes to meet a minimum requirement for their own funds and eligible liabilities, or “MREL,” and regulators vary this requirement by the size of the bank. With some minor exceptions, liabilities qualifying for MREL are the same as for TLAC: long-term, unsecured debt with at least a year of residual maturity (Benoit 2016). The primary difference is where those liabilities fall in the creditor hierarchy. For example, unlike TLAC, MREL includes some liabilities, such as senior preferred bonds, that rank *pari passu* with protected liabilities, such as common claims.⁷

As of 2022, European G-SIBs have to hold a minimum level of MREL at the Basel standard: 18% of RWAs or 6.75% of leverage exposures. So-called “top-tier banks”—non-G-SIB banks with assets exceeding EUR 100 billion and other designated banks—have to hold 13.5% of RWAs or 5% of leverage exposures in MREL, roughly 75% the level required of G-SIBs⁸ (Bank of Portugal 2020). EU regulators also have flexibility and discretion to apply an MREL standard based on a bank’s size and financial condition, meaning most G-SIBs and top-tier banks have institution-specific requirements.

Thought Experiment: TLAC as Savings for the FDIC

How much more of the losses could the failed banks’ non-depositor creditors have borne, and how much could the FDIC thus have saved, if US regulators had held them to a TLAC requirement?

Of course, it’s impossible to know what requirement regulators would have imposed on these banks if they had imposed one. We could simply have assumed that regulators would have applied the same rule that they applied to G-SIBs: TLAC equal to at least 18% of risk-weighted assets or 7.5% of leverage exposures. But we thought it’s more likely regulators would have taken a tailored approach, as they did in other areas, and applied a European-like standard.⁹

⁷ The EU, unlike the US, does not apply a minimum debt ratio to its TLAC (see Benahmed [2022, 7]). However, it does set an additional subordination requirement, called a bank’s total liabilities and own funds (TLOF), which is set at 8% of RWAs.

⁸ European authorities also designate some banks with less than EUR 100 billion in assets as other systemically important institutions, which are subject to the lower MREL standard (EBA 2023). Furthermore, European banks face additional MREL requirements due to capital buffers and supervisory add-ons; in practice, the average MREL requirement for large European non-G-SIBs was about 20% of RWA at the end of 2022, according to the Single Resolution Board (SRB 2023, 5). Practically all European banks will meet the EU’s capital conservation buffer, at 2.5%, if not an entire combined buffer requirement (CBR), which includes the capital conservation buffer, a countercyclical buffer, a systemic risk buffer, and a systemically important bank buffer (EBA 2017; ECB 2024).

⁹ Our calculations do not apply a TLOF-like standard, not only because these three banks had different corporate structures but for the more essential point that US law statutorily subordinates a bank’s other liabilities to its depositors (US Congress 2021). US regulators impose subordination requirements on

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Ultimately, the precise level of required TLAC should reflect its purpose, and non-G-SIBs would probably hold TLAC for a different purpose than would G-SIBs. The G-SIBs typically have multiple banking and investment banking subsidiaries that would need to be recapitalized—by converting TLAC into equity—in order to stay open through the holding company’s bankruptcy.

In contrast, a non-G-SIBs’ TLAC might be used to facilitate the FDIC’s most common form of resolution, known as a purchase-and-assumption transaction. In purchase-and-assumption transactions, the FDIC arranges for a bigger bank to purchase the bank and assume most of its liabilities, leaving the remaining assets and liabilities with the FDIC. This is the approach it took in these bank failures, except that it first had to create two bridge banks before it could line up buyers for SVB and Signature.

Crucially, the FDIC did impose losses on existing shareholders and creditors of the three banks. That meant total losses for the three banks’ shareholders (about USD 43 billion, based on separate calculations of the book value and market value at year-end), preferred equity holders (USD 7.3 billion), and other creditors (USD 4.7 billion). Therefore, the total bail-in for shareholders and creditors was about USD 55 billion (see Figure 10).

intermediate holding companies only, that is, the US holding companies of the US operations of foreign banks, which are not bound by US statutory subordination (Fed 2017).

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Figure 10: Costs of Resolution to DIF and Bail-in Investors

Costs of Resolution	Silicon Valley Bank	Signature Bank	First Republic Bank
<i>As of Dec. 31, 2022 (\$ millions)</i>			
Date Closed	Mar. 12, 2023	Mar. 12, 2023	May 1, 2023
Date Purchased	Mar. 27, 2023	Mar. 19, 2023	May 1, 2023
Buyer	First Citizens	NY Comm BC	JPMC
Cost to DIF			
Assets Purchased	72,000	38,400	186,000
Cost (assets + assumed liabilities)	55,500	35,900	182,600
Discount at Purchase Price	(16,500)	(2,500)	(3,400)
FDIC's Estimated Loss to DIF (≥discount)	(20,000)	(2,500)	(13,000)
Total Estimated Cost to DIF			(35,500)
Costs of Bail-in for Investors			
Common Equity: Market Value	(13,602)	(7,251)	(22,297)
Common Equity: Book Value (inc. CCB)	(16,180)	(12,006)	(14,475)
Preferred Equity: Book Value	(3,646)	-	(3,633)
Debt: Book Value	(3,379)	(571)	(779)
Total Bail-in: Market Value	(20,627)	(7,822)	(26,709)
Total			(55,158)
Total Bail-in: Book Value	(23,205)	(12,577)	(18,887)
Total			(54,669)

Note: minority interests are excluded from CET1 & AT1.

Sources: Company financial reports; FDIC press releases; YPFS analysis.

In the immediate aftermath of these failures, at the time the authors performed their analysis, the FDIC's bank-funded DIF was also depleted by an estimated USD 31.5 billion.¹⁰ Roughly half, or USD 15.8 billion, resulted from the FDIC's coverage of uninsured depositors in SVB and Signature under the systemic risk exception (SRE). The FDIC is obliged to recoup the costs of protecting uninsured depositors under this exceptional treatment through special assessments, which it is in the process of levying on 113 banks (Gruenberg 2023; Mitchell 2023). The remaining half, or USD 15.7 billion, was a loss borne by the DIF due to the discounts offered to the purchasers of the failed banks (not including any loss-sharing agreements with the buyers of the failed banks) and the estimated losses on the assets the FDIC assumed and will eventually liquidate. With a TLAC requirement, there would have been more creditors available to absorb losses, providing a first line of defense for the DIF.¹¹

¹⁰ Since the failures, the FDIC has continued to update its loss estimates and the amount attributable to the protection of uninsured depositors; see, for example, FDIC (n.d.).

¹¹ Although the DIF was designed to absorb losses in the event of bank failures, its limited capacity would prove insufficient in the event of widespread contagion, which might be aggravated by doubts about its capacity to insure losses. Therefore, regulators should be interested in minimizing the impact that any single bank failure has on the fund's resources.

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This distinction between using TLAC to keep banks open and using it solely to protect the insurance fund is apparent in the European Banking Authority's most recent report on the MREL capacity of European banks. The largest European banks plan for an open-bank resolution, in which TLAC converts to equity and the bank stays open; most smaller banks plan for a transfer to an acquiring bank, similar to a purchase-and-assumption agreement, in which TLAC is available to absorb losses (EBA 2020, 9).

TLAC Conclusions

We conclude that, had the three banks raised sufficient LTD and equity to meet a version of the TLAC standard akin to that implemented in Europe, those investors would have borne more of the losses—potentially saving about USD 13.6 billion (Figure 13, Appendix C).

Meanwhile, our understanding of the actual write-downs performed during the FDIC-brokered purchase-and-assumption transactions suggests that the fundamental logic of TLAC is sound. The FDIC forced losses on the bondholders and shareholders of these three banks, where it may once have protected them in a systemic risk exception event. Perhaps more interesting is that the FDIC performed these bail-ins both under an SRE designation (SVB and Signature) and not (First Republic).

In 2019, when Martin Gruenberg was an FDIC board member but not the chair, he said that in a resolution without a buyer, and little or no unsecured debt, the least-cost resolution would require uninsured depositors to take losses. It's reasonable, then, to view the FDIC's ex ante least-cost option for resolution as requiring a bank's investors to bear more of the losses, protecting uninsured depositors. Our analysis shows that, unsurprisingly, these banks had not issued enough LTD to bail-in, and the extent of their TLAC shortfalls provide insight into how regulators may consider applying TLAC requirements to smaller US banks.

In analyzing their LTD shortfall, we make four observations.

First, in most US banking groups, the top-tier parent company issues all of the long-term, unsecured debt, and all deposits and other short-term, unsecured debt are issued by the operating subsidiary. Therefore, TLAC-eligible senior debt must be issued by a bank holding company, which structurally subordinates it to the debt issued by the holding company's subsidiaries. This "clean holding company requirement" allows a resolution authority to bail-in the holding company's debt so that the operating subsidiaries may continue as going concerns. First Republic and Signature, however, did not operate a BHC structure like SVB, technically rendering all of their outstanding debt instruments ineligible for TLAC. If regulators apply LTD requirements to more banks, they must consider whether the banks will be required to establish a BHC structure that issues LTD. This is a question of how easy it is to resolve the entity, since by law, bail-in-able debt is statutorily subordinated to protected liabilities. Recall that after the 2018 tailoring rule, the lack of a BHC also implied that First Republic and Signature could never be subject to the LCR no matter how large they grew.

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Second, due to these organizational differences, any TLAC requirement should complement resolution strategies. As we've noted, G-SIBs need TLAC to replenish their capital, such that the operating subsidiaries may stay open while the holding company goes through bankruptcy. Smaller banks need enough TLAC to execute their resolution strategies under Title 1 of the Dodd-Frank Wall Street Reform and Consumer Protection Act of 2010, which typically includes the sale of the business to a larger buyer. In response to proposals to apply TLAC to more banks, First Republic stated that its non-BHC structure meant that it did "not present the risks and complexities with respect to financial stability or resolvability" (Roffler 2023, 4). To be sure, it was sold faster than the other two banks. But it also had less than USD 1 billion in LTD to bail in and cost the FDIC's DIF an estimated USD 13 billion.

Third, a bail-in is supposed to prevent losses to insured depositors, and in this respect, there is no fundamental difference between holding the loss-absorbing capacity in debt or equity.¹²

However, regulators will need to consider the proper debt-equity ratio in any tailored TLAC requirement. The five largest US banks not currently subject to the TLAC requirement recommended that regulators allow them to raise either debt or equity (Non-GSIB Banks 2023, 23). We expect US regulators would set a debt target at one-third the TLAC target, as the Financial Stability Board (FSB) recommends, and perhaps for good reason—in principle, debtholders are more attuned than equity investors to downside risk.¹³ The existence of both debt and equity also helps regulators clearly distinguish between going concern capital and gone concern LTD. Of course, a bank with more equity should enjoy lower costs of debt funding, too.

Fourth, ensuring a minimum amount of TLAC and LTD requires continuous monitoring by banks and their supervisors. For example, one month before SVB's fateful press release, First Republic redeemed USD 500 million of its senior debt, roughly two-thirds of its outstanding TLAC-able debt (SVB 2023c). The current US rules prevent G-SIBs from redeeming LTD if doing so would cause them to breach their TLAC or LTD requirements. In Europe, redemption of any MREL-eligible liability requires the resolution authority's approval and supervisory consultation.¹⁴ The ex ante cost of this maintenance and supervision may seem onerous until we see the ex post cost of paying out depositors and managing contagion.

A fundamental power of resolution authorities like the FDIC is the power to subordinate the claims of long-term unsecured debt to those of short-term unsecured debt. This ability to

¹² From the bank's perspective, TLAC-eligible debt is generally more expensive than other types of debt but cheaper than equity. However, the composition of debt and equity changes the bank's incentives (Mendicino, Nikolov, and Suarez 2020). More equity may reduce risk-shifting behavior, but it may also reduce returns and thus encourage risk-taking. More debt may discipline bank management since it would incentivize monitoring by creditors, and therefore may prevent the bank's insolvency long before a bail-in would be necessary, but it also affects the leverage ratio of these banks. Because preferred equity counts toward capital requirements without diluting shareholders, regulators should carefully consider how bail-in-able these instruments are.

¹³ Debt is generally easier to price, and price volatility for a bank's debt may offer regulators a market-based signal for directing supervisory resources.

¹⁴ For banks engaged in market making, the European rules grant renewable, year-long exceptions to this prior permission mechanism.

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retrospectively discriminate among creditors is designed to reduce the run incentive of depositors and other short-term debt holders.

More importantly, the resolutions of these three banks demonstrate that the FDIC is willing to bail in shareholders and creditors. Nevertheless, the DIF will bear losses above this bail-in to protect uninsured depositors. These losses are later socialized via special assessments on banks that, in turn, will seek to recoup these costs from consumers and other customers.

The question boils down to a difficult policy choice between increasing self-insurance through TLAC or increasing socialized insurance through the DIF. Bankers tend to prefer the latter: TLAC is a certain cost; and that debt and equity funding is considerably more expensive than deposit funding, and probably more expensive than any future special assessments. Regulators may lean more on TLAC, particularly for the larger banks, to avoid the risk of depleting the DIF and rendering the FDIC less able to cope with further contagion during times of financial stress.

VI. Conclusions

The post-mortem reports by the regulators of SVB and other failed banks, material loss reviews by inspectors general, and analysis by the Government Accountability Office focused almost exclusively on the failures of the banks' management and supervisors to address or even recognize the risks these banks were taking. All of these reports said little to nothing about the tailoring of banking regulation, presumably viewing the regulatory framework itself as somehow out of scope.

Our analysis points to the need for stronger banking rules to empower supervisors. Whether you believe that SVB's supervisors had no idea of the risks the bank was taking, or you believe that they understood the risks but were unable or unwilling to challenge the management of a very profitable bank in a critical industry, regulation that considered excessive interest-rate risk and liquidity risk with clear consequences could have made a difference. In this case, both the IRR-BB and the LCR standards would have identified SVB as an outlier among large US banks. If the rules had been in place, supervisors also would have had a presumption to take action.

Other countries' banking systems did not experience these types of challenges despite significant increases in central bank policy rates and quantitative tightening elsewhere following the pandemic. Despite taxpayer losses in excess of 3% of US GDP during the 1980s savings and loan crisis, the US prudential framework for banks still lacks guardrails on excessive interest-rate risk, a key risk to banks. Further, tailoring of US bank regulation means that management of liquidity risk and inclusion of interest-rate marks in bank capital differs markedly based on bank size. Finally, unconventional monetary policy remains a tool that the Fed may use again the future, even as these operations can increase asset-liability management risk at US banks through the creation and shrinkage of liquidity at the Fed and uninsured banking system deposits.

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SVB's failure resulted in official sector protection of uninsured depositors through the systemic risk exception and the Federal Reserve's creation of the Bank Term Funding Program. Following the failures of SVB, First Republic, and Signature, the losses to the DIF and the FDIC's special assessment on healthy US banks were significant and could have been lower if these banks also had been subject to TLAC requirements.

Given the public expectation of government and central bank policy intervention in a crisis and already elevated levels of US official sector debt, US bank supervision must emphasize ex ante crisis prevention against the full breadth of bank risks, including interest-rate and liquidity risk.

US bank regulation needs to include quantitative limits on interest-rate risk, liquidity risk, and TLAC for more US banks—not just the systemic US banks. Such rules are particularly important given ongoing risks related to still above-target inflation, Treasury market fragility, growing risks to investor confidence in Treasuries, and the Fed's use of unconventional monetary policy, which increases the volatility of both banking system liquidity and deposits.

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VII. References

- Bank of Portugal. 2020. “Interaction between Regulatory Minimum Requirements and Capital Buffers.” Financial Stability Report, June 2020.
<https://elischolar.library.yale.edu/ypfs-documents2/5408/>
- Basel Committee on Banking Supervision (BCBS). 2004. “Principles for the Management and Supervision of Interest Rate Risk.” Bank for International Settlements, July 2004.
<https://elischolar.library.yale.edu/ypfs-documents2/5409/>
- . 2013. “Basel III: The Liquidity Coverage Ratio and Liquidity Risk Monitoring Tools.” Bank for International Settlements, January 2013.
<https://elischolar.library.yale.edu/ypfs-documents2/5473>
- . 2014. “Basel III: The Net Stable Funding Ratio.” Bank for International Settlements, October 2014.
<https://elischolar.library.yale.edu/ypfs-documents2/5410/>
- . 2015. “TLAC Quantitative Impact Study Report.” Bank for International Settlements, November 2015.
<https://elischolar.library.yale.edu/ypfs-documents2/5411/>
- . 2016a. “Standards: Interest Rate Risk in the Banking Book.” Bank for International Settlements, April 2016.
<https://elischolar.library.yale.edu/ypfs-documents2/5412/>
- . 2016b. “Standard: TLAC Holdings.” Bank for International Settlements, October 2016.
<https://elischolar.library.yale.edu/ypfs-documents2/5413/>
- . 2025. “BCBS Dashboards.” Bank for International Settlements, 2025.
<https://elischolar.library.yale.edu/ypfs-documents2/5402/>
- Benahmed, Riad. 2022. “Loss-Absorbing Capacity Requirements in Resolution for G-SIBs in the EU and the US: A Cross-Jurisdictional Comparison.” *ACPR, Bank of France, Working Paper on Resolution*, December.
<https://elischolar.library.yale.edu/ypfs-documents2/5414/>
- Benoit, Mesnard. 2016. “Loss Absorbing Capacity in the Banking Union: TLAC Implementation and MREL Review,” July 7, 2016.
<https://elischolar.library.yale.edu/ypfs-documents2/5415/>
- Board of Governors of the Federal Reserve System (Fed). 1996. “FRB: Supervisory Letter on Joint Policy Statement on Interest Rate Risk,” May 23, 1996.
<https://elischolar.library.yale.edu/ypfs-documents2/5417/>

POLICY NOTE

———. 2017. “Total Loss-Absorbing Capacity, Long-Term Debt, and Clean Holding Company Requirements for Systemically Important U.S. Bank Holding Companies and Intermediate Holding Companies of Systemically Important Foreign Banking Organizations.” 82 Fed. Reg. 8266, January 24, 2017.

<https://elischolar.library.yale.edu/ypfs-documents2/5431/>

———. 2019. “Requirements for Domestic and Foreign Banking Organizations.” Tailoring Rule, 2019.

<https://elischolar.library.yale.edu/ypfs-documents2/5432/>

———. 2020. “Final Rule to Implement a Net Stable Funding Ratio Requirement for Large Banking Organizations.” Draft Final Rule, October 20, 2020.

<https://elischolar.library.yale.edu/ypfs-documents2/5450/>

———. 2023a. “Re: Review of the Federal Reserve’s Supervision and Regulation of Silicon Valley Bank,” April 28, 2023.

<https://elischolar.library.yale.edu/ypfs-documents2/4025/>

———. 2023b. “Section 3300.1: Interest Risk Management.” In *Commercial Bank Examination Manual*. Washington, DC: The Federal Reserve.

<https://elischolar.library.yale.edu/ypfs-documents2/5418/>

———. 2024. “12 CFR Part 249 Subpart D -- Total Net Cash Outflow.” National Archives, December 12, 2024.

<https://elischolar.library.yale.edu/ypfs-documents2/5419/>

———. 2025. “Capital Adequacy of Bank Holding Companies, Savings and Loan Holding Companies, and State Member Banks (Regulation Q).” 12 CFR Part 217, January 22, 2025.

<https://elischolar.library.yale.edu/ypfs-documents2/5451/>

Board of Governors of the Federal Reserve System and Federal Deposit Insurance Corporation (Fed and FDIC). 2022. “Resolution-Related Resource Requirements for Large Banking Organizations.” 87 Fed. Reg. 64170. R-1786, October 24, 2022.

<https://elischolar.library.yale.edu/ypfs-documents2/5420/>

Board of Governors of the Federal Reserve System, Federal Deposit Insurance Corporation, and Office of the Comptroller of the Currency (Fed, FDIC, and OCC). 2014. “Liquidity Coverage Ratio: Liquidity Risk Measurement Standards.” 79 Fed. Reg. 61440, October 10, 2014.

<https://elischolar.library.yale.edu/ypfs-documents/13123/>

———. 2019. “Changes to Applicability Thresholds for Regulatory Capital and Liquidity Requirements.” 84 Fed. Reg. 59230, November 1, 2019.

<https://elischolar.library.yale.edu/ypfs-documents2/5445/>

POLICY NOTE

———. 2020. “Agencies Issue Final Rule to Strengthen Resilience of Large Banks.” Joint press release, October 20, 2020.
<https://elischolar.library.yale.edu/ypfs-documents2/5416/>

———. 2023. “Long-Term Debt Requirements for Large Bank Holding Companies, Certain Intermediate Holding Companies of Foreign Banking Organizations, and Large Insured Depository Institutions.” 88 Fed. Reg. 64524, September 19, 2023.
<https://elischolar.library.yale.edu/ypfs-documents2/5497>

Capital One Financial Corporation, Charles Schwab Corporation, PNC Financial Services Group, Truist Financial Corporation, and U.S. Bancorp (Non-G-SIB Banks). 2023. “Comment Letter to the Board of Governors of the Federal Reserve System and Federal Deposit Insurance Corporation,” January 23, 2023.
<https://elischolar.library.yale.edu/ypfs-documents2/5545>

Cetina, Jill. 2015. “Incorporating Liquidity Shocks and Feedbacks in Bank Stress Tests.” *OFR Brief Series* 15, no. 6: 7.
<https://elischolar.library.yale.edu/ypfs-documents2/5421/>

Cipriani, Marco, Thomas M. Eisenbach, and Anna Kovner. 2024. “Tracing Bank Runs in Real Time.” Federal Reserve Bank of New York Staff Report No. 1104, September 2024.
<https://elischolar.library.yale.edu/ypfs-documents2/4763/>

Cleary Gottlieb Steen & Hamilton LLP (Cleary Gottlieb). 2015. “Comparing the Federal Reserve’s TLAC Proposal to the FSB TLAC Final Standards.” Alert Memorandum, November 20, 2015.
<https://elischolar.library.yale.edu/ypfs-documents2/5422/>

European Banking Authority (EBA). 2017. “2015_1759 Group Solvency - Combined Buffer Requirements.” Single Rulebook Q&A, July 28, 2017.
<https://elischolar.library.yale.edu/ypfs-documents2/5424/>

———. 2020. “EBA Quantitative MREL Report.” EBA/REP/2022/09. EBA Reports, December 31, 2020.
<https://elischolar.library.yale.edu/ypfs-documents2/5425/>

———. 2023. “Other Systemically Important Institutions (O-SIIs).” Risk Monitoring, June 12, 2023.
<https://elischolar.library.yale.edu/ypfs-documents2/5426/>

European Central Bank (ECB). 2024. “Capital Conservation Buffer.” European Systemic Risk Board, March 20, 2024.
<https://elischolar.library.yale.edu/ypfs-documents2/5427/>

POLICY NOTE

Federal Deposit Insurance Corporation (FDIC). n.d. "Special Assessment Pursuant to Systemic Risk Determination." Accessed January 24, 2025.
<https://elischolar.library.yale.edu/ypfs-documents2/5429/>

Financial Stability Board (FSB). 2015. "Principles on Loss-Absorbing and Recapitalisation Capacity of G-SIBs in Resolution: Total Loss-Absorbing Capacity (TLAC) Term Sheet," November 9, 2015.
<https://elischolar.library.yale.edu/ypfs-documents2/5434/>

First Republic Bank (First Republic). 2023. "Form 10-K." Press release, February 13, 2023.
<https://elischolar.library.yale.edu/ypfs-documents2/5202>

Gaul, Lewis, and Jonathan Jones. 2021. "CAMELS Ratings and Their Information Content." OCC Working Paper No. WP-2021-01, January 6, 2021.
<https://elischolar.library.yale.edu/ypfs-documents2/5470>

Gopalan, Yadav, and Joao Granja. 2023. "How (In)Effective Was Bank Supervision During the 2022 Monetary Tightening?" University of Chicago, Becker Friedman Institute for Economics Working Paper No. 2023-130, September 29, 2023.
<https://elischolar.library.yale.edu/ypfs-documents2/5472>

Gruenberg, Martin J. 2023. "Oversight of Financial Regulators: Financial Stability, Supervision, and Consumer Protection in the Wake of Recent Bank Failures." Testimony of the FDIC Chairman before the US Senate Committee on Banking, Housing, and Urban Affairs, May 18, 2023. <https://elischolar.library.yale.edu/ypfs-documents2/5428/>

Honohan, Patrick. 2023. "The Fed's Recent Bank Stress Tests Used Scenarios with Little Variation. None Examined Higher Interest Rates." The Peterson Institute for International Economics, March 27, 2023.
<https://elischolar.library.yale.edu/ypfs-documents2/5437/>

Mendicino, Caterina, Kalin Nikolov, and Javier Suarez. 2020. "Equity Versus Bail-In Debt in Banking: An Agency Perspective." ESRB Working Paper Series, No. 2017/50, July 2020.
<https://www.ssrn.com/abstract=3723397>

Mitchell, Patrick. 2023. "Re: Notice of Proposed Rulemaking on Special Assessments Pursuant to Systemic Risk Determination." Federal Deposit Insurance Corporation, May 11, 2023.
<https://elischolar.library.yale.edu/ypfs-documents2/5438/>

Office of Financial Research (OFR). 2015. Annual Report 2014.
<https://elischolar.library.yale.edu/ypfs-documents2/5443/>

POLICY NOTE

Office of the Comptroller of the Currency (OCC). 2020. "Interest Rate Risk (Version 1.0)." In *Comptroller's Handbook*. Safety and Soundness Series. Washington, D.C.: Office of the Comptroller of the Currency.

<https://elischolar.library.yale.edu/ypfs-documents2/5455/>

———. 2024. "Interest Rate Risk Statistics Report [Fall 2024]," October 21, 2024.

<https://elischolar.library.yale.edu/ypfs-documents2/5442/>

Roffler, Michael J. 2023. "Comment Letter to the Board of Governors of the Federal Reserve System and Federal Deposit Insurance Corporation," January 23, 2023.

<https://elischolar.library.yale.edu/ypfs-documents2/5435/>

Rossolillo, Nicholas. 2021. "SVB Financial Group Is the Best Bank Stock Nobody's Talking About." *Motley Fool*, May 16, 2021.

<https://elischolar.library.yale.edu/ypfs-documents2/5407/>

Silicon Valley Bank Financial Group (SVB). 2022. "Systemic Risk Report—FR Y-15 [SVB Financial Group]," December 31, 2022.

<https://elischolar.library.yale.edu/ypfs-documents2/5423/>

———. 2023a. "SVB Q4 2022 Financial Highlights," January 19, 2023.

<https://elischolar.library.yale.edu/ypfs-documents2/5448/>

———. 2023b. "Form 10-K," February 24, 2023.

<https://elischolar.library.yale.edu/ypfs-documents2/5447/>

———. 2023c. "SVB Financial Group Announces Proposed Offerings of Common Stock and Mandatory Convertible Preferred Stock." Press release, March 8, 2023.

<https://elischolar.library.yale.edu/ypfs-documents2/5546/>

Single Resolution Board (SRB). 2023. "SRB MREL Dashboard Q4.2022," May 15, 2023.

<https://elischolar.library.yale.edu/ypfs-documents2/5449/>

US Congress. 2021. "Section 5. Deposit Insurance." Federal Deposit Insurance Act, August 31, 2021. 12 CFR Part 330, August 31, 2021.

<https://elischolar.library.yale.edu/ypfs-documents2/5430/>

US Securities and Exchange Commission (US SEC). 1997. "Questions and Answers about the New 'Market Risk' Disclosure Rules," July 31, 1997.

<https://elischolar.library.yale.edu/ypfs-documents2/5454/>

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VIII. Appendixes

Appendix A: LCR Calculations for SVB

Appendix A describes our liquidity coverage ratio (LCR) calculations for Silicon Valley Bank (SVB) (Figure 11). The caveat is that we were limited to public reports—the company’s 10-K, Y-15 systemic risk report, call reports, and investor presentations. The Federal Reserve did require the company to submit the FR 2052A liquidity monitoring report, but that is not publicly available. We believe we made the most of the information that was publicly available.¹⁵

¹⁵ Note that this piece focuses on SVB. A similar analysis of First Republic Bank suggests it also would have been out of compliance with the LCR if it had been required to meet it. Its Form 10-K for 2022 notes that, although the LCR didn’t apply, the bank had USD 26 billion in high-quality liquid assets (HQLAs). But that would be its HQLA ignoring the haircuts and cap on Level 2 assets that the LCR rule imposes. Based on the rule, its HQLA would have been about USD 10 billion. Meanwhile, First Republic had USD 176 billion in deposits, of which about one-third was insured; USD 51 billion in commitments; USD 10 billion in short-term Federal Home Loan Bank commitments; and minimal inflows. Assuming a 3% drawdown for insured deposits and 10% for uninsured deposits (generously) gets an LCR of roughly 50%, before the 70% adjustment for banks with less than USD 250 billion in assets (First Republic 2023).

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Figure 11: LCR Calculations

Numerator: High Quality Liquid Assets			
\$ billions		Haircut	Contribution
Level 1			
Reserve Balances	7.8	0%	7.8
US Treasuries, Sovereign Debt	16.2	0%	16.2
US Government: Ginnie Mae	7.7	0%	7.7
Total			31.7
Level 2A			
Agency Debt and MBS	70.0	15%	59.5
Total (capped at 40% of HQLA)			21.2
Total HQLA			52.9

Denominator: Cash Outflows				
\$ billions		Drawdown Min-Max	Drawdown Used Here	Contribution
Outflows				
Total Deposits	173.1			
Insured Deposits	7.7	3-40%	5%	0.4
Uninsured Deposits (A)	165.4			
Nonfinancial wholesale (Y-15) (B)	109.6			
Operational, noninterest-bearing: 47%	51.5	25%	25%	12.9
Non-operational, interest-bearing: 53%	58.1	40%	40%	23.2
Financial wholesale (Y-15)	28.8			
Operational, noninterest-bearing: 47%	13.5	25%	25%	3.4
Non-operational, interest-bearing: 53%	15.3	100%	100%	15.3
Uninsured retail or small business deposit	27.1	10%	10%	2.7
Short-term Borrowing (mostly FHLBs)	13.6	0-30%	25%	3.4
Commitments	62.5	5-40%	20%	12.5
Total Outflow				73.7

Note: Most figures are from SVB's Form 10-K or Form Y-15 for 2022.

(A) "Uninsured deposits in U.S. offices that exceed the FDIC insurance limit were \$151.5 billion and... foreign deposits were \$13.9 billion" at the end of 2022. SVB 10-K, 2022.

(B) The Y-15 shows SVB's estimate of wholesale funding, with financial and nonfinancial breakdowns. We assume the remainder is retail-small business.

(C) Nonfinancial business represents the sum of client funds in technology or health science clients, per SVB Investor Presentation, January 2023.

Denominator: Cash Inflows			
Deposits at Financial Institutions	2.6	100%	2.6
Total Loans	73.6		
Revolving Loans	46.0	0%	-
Investor-dependent loans	6.1	0%	-
Other loans: More than one year	18.6	0%	-
Other loans: Less than one year (par)	2.9		
Assume 1/12	0.2	50%	0.1
Total Inflow	-		2.8

LCR Calculation	
Net Cash Outflow	71.0
Liquidity Coverage Ratio	75%
Liquidity shortfall @ 100% LCR	18.1
Liquidity Shortfall @ 125% LCR	35.8

Sources: Company financial reports; YPFS analysis.

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LCR Numerator: High-Quality Liquid Assets

High-quality liquid assets (HQLAs) were relatively easy to calculate. Level 1 assets are considered the most liquid assets and enter the numerator with no haircut. They include reserve balances with the central bank, government debt, and government-guaranteed securities. At the end of 2022, SVB had USD 31.7 billion in Level 1 assets: USD 7.8 billion in reserve balances; USD 16.2 billion in US Treasuries; and USD 7.7 billion in mortgage securities issued by Ginnie Mae, which the government fully guarantees. Level 2A assets are considered somewhat riskier and receive a 15% haircut. They include government agency debt and agency mortgage-backed securities. SVB had more than USD 60 billion in agency mortgage-backed securities. However, under the LCR rules, Level 2A assets can't be more than 40% of total HQLA, so SVB's Level 2A assets would have been capped at USD 21.2 billion.

Total HQLA: USD 52.9 billion.

LCR Denominator: Cash Outflows

The denominator is harder to calculate and open to some interpretation. If SVB had been subject to the rule, its management would have had to hammer out all of the issues in the following paragraphs with its Fed supervisors, using far more detailed data disclosures than the company was required to make to the public.

To compute outflows, the LCR rules assume stressed runoff rates for each type of bank liability based on experience in past crises and on management and supervisory judgment. SVB had USD 173.1 billion in deposits at the end of 2022. Of those, USD 165.4 billion were uninsured: USD 151.5 billion were uninsured deposits in US offices that exceeded the Federal Deposit Insurance Corporation (FDIC) insurance limit; another USD 13.9 billion were foreign deposits, which the FDIC doesn't insure (see the company's last Form 10-K [SVB 2023b]). We assumed the remaining USD 7.7 billion was insured. Under the US rules, stressed outflows for insured deposits range from 3% to 40%: 3% for retail and small business customers, and 5% to 40% for wholesale customers that are not financial companies. We assumed an average 5% runoff of insured deposits. That yielded a small outflow of USD 0.4 billion.

Calculating outflows on the USD 165.4 billion in *uninsured* deposits was the largest part of SVB's LCR and required the most judgment. These were also the depositors who ran the bank, so it is important to understand how the rule would have treated them *ex ante*.

Within the uninsured deposit category, the LCR rules makes key distinctions:

- Retail clients are considered more loyal (sticky) and less likely to run than wholesale clients.
- Among wholesale accounts, operational deposits—deposits that customers need to place with a bank in order to use services such as payment settlement systems, and don't bear interest—are considered stickier than nonoperational deposits—which

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are interest bearing; the rules assume these depositors are more likely to switch to other banks to get better rates.

- Also among wholesale accounts, nonfinancial company deposits are considered stickier than financial company deposits.

SVB reported its end-2022 short-term wholesale funding in its Y-15 Systemic Risk Report, as the Fed requires (SVB 2022). Uninsured, nonfinancial wholesale deposits were USD 109.6 billion; uninsured, financial wholesale deposits were USD 28.8 billion. It is typically difficult to estimate from public filings how much of a bank's deposits are operational, but the company in a January 2023 investor presentation said that 47% of clients' funds were in "operating cash typically held in on-balance sheet, noninterest-bearing deposits" (SVB 2023a, 11). Accepting the company's definition of "operating" and applying that proportion yielded USD 65 billion in operational uninsured deposits from both financial and nonfinancial customers, which have a 25% runoff factor under the rule; USD 58.1 billion in nonoperational uninsured deposits from nonfinancial customers (40%); and USD 15.3 billion in nonoperational uninsured deposits from financial customers (100%). The combined outflow from uninsured wholesale clients came to USD 54.7 billion.

We assumed the remaining USD 27.1 billion of uninsured deposits were retail or small business. Applying the 10% runoff factor under the rule yielded an outflow of USD 2.7 billion.

The LCR rules also require a bank to assume that its customers draw down on a portion of their outstanding lines of credit and other commitments. The drawdown rates are 5% for commitments to retail customers, 10% or 30% for nonfinancial wholesale customers, and 40% for financial entities other than banks (Fed 2024, part 249.32 [e][1]). SVB had USD 62.2 billion in such commitments at the end of 2022; assuming a 20% average drawdown rate, the outflow would have been USD 12.5 billion.

Finally, SVB had USD 13.6 billion in short-term borrowings, of which USD 13 billion were advances from the Federal Home Loan Bank (FHLB) of San Francisco. The outflow for such borrowings was 25%, or USD 3.4 billion.

The total cash outflow came to USD 73.7 billion.

LCR Denominator: Cash Inflows

Next, inflows. The LCR allows banks to subtract contractual inflows from their estimate of 30-day outflows, according to the LCR guidelines (BCBS 2013, para. 142).

Contractual inflows may include loans that are scheduled to mature during the 30-day period. The rules then allow banks to treat half of such repayments as inflows. However, a close look at SVB's USD 73.6 billion portfolio of performing loans suggests that it should have counted on very little of those loans to be repaid over 30 days.

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First, of that total, USD 46 billion were revolving loans (SVB 2023b, 132–33). Borrowers are expected to roll over such loans, and they are not included in inflows, per the US final rule: “With respect to revolving credit facilities, already drawn amounts would not have been included in a covered company’s inflow amount” (Fed, FDIC, and OCC 2014). To make clear why such loans should not be considered inflows in SVB’s case, note that about USD 40 billion of the total represented loans to private equity or venture capital funds. According to the company: “The vast majority of this portfolio consists of capital call lines of credit, the repayment of which is dependent on the payment of capital calls by the underlying limited partner investors in the funds managed by these firms” (SVB 2023b, 70). Surely, it makes sense that the rules would not expect SVB to be able to demand that venture capital investors repay their revolving loans in stressed market conditions.

Eliminating the revolving loans left USD 27.6 billion in term loans. These included USD 6.1 billion in “investor-dependent” loans. Borrowers of these loans may have required additional equity financing from venture capital firms or other investors, or in some cases a successful sale to a third party or an IPO, according to the company’s 10-K (SVB 2023b, 70). Further, the risk of “prolonged market volatility... particularly applies to Investor Dependent loans, where repayment is dependent on the borrower’s ability to fundraise or exit” (74). Without specific backing from the Basel Committee on Banking Supervision guidelines or the agencies’ final rule, it wouldn’t have been prudent to assume repayment of investor-dependent loans in stressed market conditions. Of the remaining USD 21.5 billion in term loans, less than USD 3 billion would have matured in under one year (SVB 2023b, 73). Assuming one-twelfth of those loans paid back at maturity within 30 days and applying the 50% haircut in the rule gave a cash inflow of USD 0.1 billion.

SVB reported USD 5.3 billion in deposits at other financial institutions. Under the rules, banks can’t include operational deposits in their inflow estimates, since “these deposits are required for operational reasons, and are therefore not available to the depositing bank to repay other outflows” (BCBS 2013, para. 98). The US regulators’ final rule in 2014 similarly excluded “any inflows derived from amounts that a covered company holds in operational deposits at other regulated financial companies... [T]he agencies reasoned that it would be unlikely that a covered company would be able to withdraw these funds in a crisis to meet other liquidity needs, and therefore excluded them” (Fed, FDIC, and OCC 2014). However, it’s impossible to tell from the public filings how much of SVB’s bank deposits were operational. We assumed 50% were operational. That yielded an inflow of USD 2.6 billion, using the 100% factor in the rule.

Cash inflows: USD 2.8 billion. Net outflows: USD 71 billion.

The resulting LCR is 75%: 52.9/71.

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Appendix B: NSFR Calculations for SVB

As in the liquidity coverage ratio (LCR) calculations, we made the most of publicly available information. Interpretations of the rule were largely based on the memo that Federal Reserve Board staff published with the press release in 2020 (Fed 2020). The company may have reported different figures if supervisors had required it to report its net stable funding ratio (NSFR).

The NSFR applies multipliers, called factors, to reflect the relative stability of each funding source in the numerator and the relative long-term nature of each asset that needs funding in the denominator. For example, in the numerator, capital and long-term debt are considered 100% stable and insured retail deposits are considered 95% stable; the more of these liabilities, the better. Uninsured wholesale deposits get a much lower stable funding factor of 0% to 50%, depending on nuances that we discuss below.

In the denominator, loans with more than one year left to mature typically require a lot of stable funding. Their required stable funding factors range from 65% for relatively low-risk, nonfinancial loans to 100% for long-term financial loans.

Most *securities*, however, are considered short-term holdings and require far less stable funding. The most liquid securities, which regulators consider to be Level 1 high-quality liquid assets (HQLAs)—central bank reserves, US Treasuries, and other government-backed securities—don't require any stable funding at all. Level 2A HQLAs require 15% stable funding. These include agency mortgage-backed securities (MBS) issued by the government-sponsored enterprises, Fannie Mae and Freddie Mac, which dominated Silicon Valley Bank's (SVB's) portfolio.

NSFR Numerator: Available Stable Funding

SVB had USD 17.9 billion in regulatory capital and USD 5.4 billion in long-term debt at the end of 2022, contributing a total of USD 23.3 billion in available stable funding based on the 100% funding factor in the rule. Insured retail deposits of USD 7.7 billion contributed USD 7.3 billion (95%); uninsured small business and retail deposits of USD 27.1 billion, based on our estimates, contributed USD 24.4 billion (90%).

The bank also reported USD 13 billion in short-term funding from the Federal Home Loan Banks. The rule (Fed, FDIC, and OCC 2020, 105) attributes a 50% available stable funding factor to such debt, resulting in a contribution of USD 6.5 billion.

The rule appropriately considers uninsured wholesale deposits to be far less stable than insured deposits. And it considers deposits that customers need for their operations (for example, payment settlement) to be more stable than other deposits. Operational deposits from financial or nonfinancial companies receive a 50% funding factor. Nonoperational deposits receive a 50% factor if they are from nonfinancial companies and a 0% factor if they are from financial companies.

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As with the LCR, one of the challenges in estimating NSFRs with only public information was the allocation of uninsured wholesale deposits between the operational and nonoperational categories. Again accepting the company's definition of "operating" from its January 2023 investor presentation suggested a 47%/53% breakdown between operational and nonoperational deposits (SVB 2023a). These assumptions yielded available funding of USD 32.5 billion from operational deposits and USD 29 billion from nonoperational deposits held by nonfinancial companies.

Total available stable funding: USD 123 billion.

NSFR Denominator: Required Stable Funding

As noted, the NSFR rules are generous for securities that qualify as HQLAs, because they assume repurchase agreement (repo) funding availability. And the NSFR—like the LCR, in fact—doesn't distinguish between held-to-maturity securities and securities that a bank holds for sale or trading. Thus, the NSFR rule would have treated SVB's agency MBS as Level 2A HQLAs with a required stable funding factor of just 15% of their carrying value.

The bank held USD 85.1 billion in agency MBS. Of that, USD 2 billion were pledged to the Federal Home Loan Banks for more than one year; the rule considers such encumbered securities to be unavailable, so they required 100% stable funding. The remaining USD 83.1 billion—of which 97% was held-to-maturity—would have required just USD 12.5 billion in stable funding.

The bank held an additional USD 8.1 billion of Level 2B HQLAs, like municipal bonds, also held-to-maturity. They carried a 50% required stable funding factor, so they would have required USD 4.1 billion in stable funding.

SVB had USD 62.5 billion in undrawn commitments to provide credit or liquidity to its clients. These carried a 5% funding factor, resulting in USD 3.1 billion in required stable funding. We also assumed USD 3 billion in required stable funding to back SVB's deposits with other banks and USD 2.7 billion to back nonmarketable securities.

Loans would have contributed the bulk of SVB's required stable funding, even though they made up only one-third of the bank's balance sheet. More than half of the bank's loans were revolving loans (USD 40.5 billion total) from SVB's Global Fund Banking business line. We assumed these would have had a 100% required funding factor in the rule, assuming clients had the ability to roll them over indefinitely; relaxing that assumption could have reduced required funding by about USD 20 billion. The bank's 2022 Form 10-K noted that repayment of much of its loans were dependent upon clients' receipt of funding from third parties (SVB 2023b, 19). During stressed conditions, it seems fair to say that SVB couldn't have been sure when clients would have repaid these loans.

There was another USD 5.5 billion in revolving loans and USD 18.4 billion in other loans to nonfinancial institutions, mostly carrying an 85% required funding factor. These would have required USD 4.7 billion and USD 14.2 billion in stable funding, respectively. A small amount

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of non-revolving financial loans would have required USD 0.4 billion. Residential mortgages (USD 9.1 billion) would have required USD 5.9 billion, based on a 65% risk factor.

Total required funding for loans: USD 65.8 billion.

Total required stable funding: USD 93.1 billion.

The resulting NSFR is 132%: 123/93.1.

Figure 12: NSFR Calculations

Numerator: Available Stable Funding			
<i>\$ billions</i>		Funding	
		Factor	Contribution
Regulatory Capital	17.9	100%	17.9
Liabilities > one year	5.4	100%	5.4
FHLB Liabilities < one year	13.5	50%	6.5
Insured Deposits	7.7	95%	7.3
Uninsured Retail/Small Business Deposits	27.1	90%	24.4
Operational Deposits (non-interest-bearing)	65.0	50%	32.5
Non-op, Non-financial Deposits (interest-bearing) *	58.1	50%	29.0
Non-op, Financial Deposits (interest-bearing) **	15.3	0%	0.0
Total Available Stable Funding			123.0

Denominator: Required Stable Funding			
<i>\$ billions</i>		Funding	
		Factor	Contribution
Level 1 HQLA			
Reserve balances	7.8	0%	0.0
US Treasuries, Foreign Sovereign Debt	18.5	0%	0.0
US Government: Ginnie Mae	7.7	0%	0.0
Level 2A HQLA (of which HTM: \$82.7 billion)			
Pledged to FHLB > one year	2.0	100%	2.0
Not Pledged	83.1	15%	12.5
Level 2B HQLA (all HTM)	8.1	50%	4.1
Undrawn Commitments	62.5	5%	3.1
Operational deposits with banks	6.0	50%	3.0
Nonmarketable securities	2.7	100%	2.7
Total Loans	74.3		
Revolving loans ***	46.0	85-100%	45.2
Mortgages	9.1	65%	5.9
Other loans to nonfinancial institutions	18.4	50-85%	14.2
Other loans to financial institutions	0.7	50-100%	0.4
Total Required Stable Funding			93.1

NSFR	132%
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* Assume all <1 year

** Assume all <6 months

*** Assume revolving loans >1 year

Sources: Company financial reports; YPFS analysis.

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Appendix C: TLAC Calculations for SVB and Other Failed Banks

We sought to calculate whether the Basel Committee on Banking Supervision's existing total loss-absorbing capacity (TLAC) rules, as tailored by European regulators, could have saved the Federal Deposit Insurance Corporation's Deposit Insurance Fund money by allowing it to bail in more long-term debt (LTD) creditors. In short, our analysis showed that the three failed banks did not have sufficient LTD at the level or composition that regulators would have required for an effective bail-in.

Not all LTD can effectively absorb losses; following the Financial Stability Board's 2015 recommendations, regulators require any TLAC debt to be unsecured and subordinated to depositors and other protected creditors (FSB 2015). Seeking a simple and straightforward bail-in, US regulators in their implementation of the Basel standard have gone a step further by requiring debt instruments to be "plain vanilla," that is, without convertibility or derivative-linked features, such as call options. European Union (EU) regulators, on the other hand, are more generous in what qualifies as TLAC, accepting more Tier 2 instruments, structured notes, and other convertible regulatory capital instruments, which the US does not accept¹⁶ (Cleary Gottlieb 2015).

Of course, TLAC would be one requirement of many already applied by regulators. For example, these three banks were also subject to a capital conservation buffer of 2.5% of their risk-weighted assets (RWAs), which they had to meet entirely with common equity Tier 1 (CET1) capital (Fed 2025). If they breached their minimum capital requirements plus this buffer (that is, a total CET1 ratio of 7%), regulators would force the banks to pause capital distributions. To ensure that TLAC is unencumbered, banks cannot count CET1 for both their required buffers and their TLAC requirement. (This is also true of global systemically important banks (G-SIBs), which are subject to more and larger buffers [Fed 2017].) For our calculation, we earmarked 2.5% of the banks' RWAs to meet this buffer, meaning it could not count toward TLAC (BCBS 2016b).

Our analysis applied the EU's threshold for similarly sized banks, which is 13.5% of risk-weighted assets.¹⁷ We also applied the US rules for eligible instruments and debt-equity composition. That is, we expected an LTD target of 4.5% of RWAs, or one-third of the 13.5% TLAC target, and an equity target of 9% of RWAs.

Based on our calculations, these three US banks would have had a combined equity and LTD shortfall of USD 13.6 billion under these assumptions:

¹⁶ There are many differences in the US and EU rules. For example, regulators in the US and EU only count LTD with a residual maturity of at least one year, but the US goes further by haircutting eligible LTD by 50% if it matures in less than *two* years (Cleary Gottlieb 2015). This is a material difference in rulemaking, but for the purposes of our analysis, we left this issue aside, since all the eligible LTD issued by these three banks had remaining maturities of greater than two years.

¹⁷ We did not use the leverage exposure measure since these banks, unlike G-SIBs, were not subject to consistent leverage reporting requirements.

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Silicon Valley Bank. At year-end 2022, Silicon Valley Bank's (SVB's) liabilities included enough TLAC-eligible equity and debt instruments to exceed the EU's minimum 13.5% threshold. However, those instruments were mostly CET1 or preferred stock, and long-term debt made up just 3% of RWAs. To meet that part of the US standards, SVB would have had to raise an additional USD 1.7 billion in LTD or convert some of its excess capital into LTD.

Signature Bank. At the same point in time, Signature's TLAC ratio was only 7.1%, substantively all of which was in CET1. To comply with a 13.5% TLAC threshold and a 4.5% LTD threshold, Signature would have had to raise another USD 3.5 billion in LTD. On the capital side, we assumed Signature would set aside enough CET1 to continue its capital distributions. Excluding the CET1 that would have met the capital conservation buffer (2.5% of RWA) would have placed Signature in a TLAC-capital deficit of USD 2.3 billion. The combined shortfall was USD 5.8 billion.

First Republic. First Republic had only USD 779 million in long-term senior notes outstanding, and a TLAC ratio of 9.5% at year-end 2022. To meet our hypothetical requirements, First Republic would have had to raise an additional USD 6.1 billion in LTD.

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Figure 13: TLAC Deficit Calculations

Total Loss-Absorbing Capital (TLAC)	Silicon Valley Bank	Signature Bank	First Republic Bank
<i>As of Dec. 31, 2022 (\$ millions)</i>			
Total Assets	211,800	110,400	212,600
Total Deposits	173,109	88,589	176,437
Deposits, % of total liabilities	89%	87%	90%
% Uninsured deposits	88%	90%	68%
Tier 1 Capital			
<i>Common Equity Tier 1 (CET1)</i>			
Common stock + paid-in capital	5,318	4,552	6,258
Retained earnings	8,951	5,457	7,886
Less: Adj other comprehensive income	(1,911)	(1,997)	(331)
Less: Conservation buffer (2.5% of RWA)	(2,841)	(2,245)	(3,794)
<i>Additional Tier 1 (AT1)</i>			
Preferred stock	3,646	-	3,633
Total TLAC-eligible Capital (Book value)	13,163	5,767	13,652
Eligible external long-term debt (LTD)			
<i>Unsecured Debt</i>			
Senior	3,288	-	779
Subordinated	91	571	-
Total TLAC-eligible Debt	3,379	571	779
TLAC-eligible debt as % of total LTD	63%	79%	61%
TLAC Deficit Calculations	Silicon Valley Bank	Signature Bank	First Republic Bank
<i>As of Dec. 31, 2022 (\$ millions)</i>			
Total TLAC (Tier 1 Capital + LTD)	16,542	6,338	14,431
Equity % of TLAC	80%	91%	95%
Debt % of TLAC	20%	9%	5%
Risk-weighted assets (RWA)	113,628	89,794	151,777
TLAC Ratio	14.6%	7.1%	9.5%
<i>Applying the tailored European rule</i>			
Minimum TLAC @ 13.5% of RWA	15,340	12,122	20,490
TLAC Surplus (Deficit) @ 13.5%	1,203	(5,784)	(6,059)
Minimum Capital @ 9.0% of RWA	10,227	8,081	13,660
Capital Surplus (Deficit) @ 9%	2,937	(2,315)	(8)
Minimum LTD @ 4.5% of RWA	5,113	4,041	6,830
LTD Surplus (Deficit) @ 4.5%	(1,734)	(3,470)	(6,051)
Sum of Capital and LTD Deficits @ 13.5%			(13,578)
<i>Applying the Basel G-SIB rule</i>			
Minimum TLAC @ 18.0% of RWA	20,453	16,163	27,320
TLAC Surplus (Deficit) @ 18.0%	(3,911)	(9,825)	(12,889)
Minimum Capital @ 12.0% of RWA	13,635	10,775	18,213
Capital Surplus (Deficit) @ 13.5%	(472)	(5,008)	(4,562)
Minimum LTD @ 6.0% of RWA	6,818	5,388	9,107
LTD Surplus (Deficit) @ 18.0%	(3,439)	(4,817)	(8,328)
Sum of Capital and LTD Deficits @ 18%		Total:	(26,625)
Note: minority interests are excluded from CET1 & AT1.			
Sources: Dec. 31, 2022 Call Reports, 10-Ks, FDIC press releases, Refinitiv.			

Sources: Company financial reports; YPFS analysis.

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